# When does tail-risk optimization change a quantum gate? An equivalence boundary for mean and CVaR control

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Robust quantum control usually minimizes the *mean* gate error over calibration noise, although fault tolerance is limited by rare bad gates. Tail-risk objectives such as the conditional value-at-risk (CVaR) have been proposed, but it is not known when they actually select a different pulse. We prove that when the noise is Gaussian and enters the gate error linearly, the error distributions of all controls are rescaled copies of one shape. All controls are then ordered by stochastic dominance, and the mean, CVaR and every other law-invariant monotone risk measure share one minimizer, up to an explicit residual from the periodicity of the infidelity. The ordering survives temporally correlated noise and several channels on one axis, and fails for channels on different axes. Simulations of a transmon $X_{\pi/2}$ gate, with full Lindblad re-evaluation, confirm the equivalence up to an accumulated detuning phase variance of about 0.05 at three gate times. Over pools of about 400 candidate pulses per setting, not one of 19 639 well-separated pairs is ranked differently by the two objectives, and the predicted tail-to-mean ratio of 5.58 holds to 2%. Leakage does not break the equivalence. Beyond the boundary, CVaR lowers the worst-5% error by up to 37% by deliberately miscalibrating the gate to pre-compensate the second-order error of the worst noise draws. The ratio $\kappa$ of second- to first-order error orders every divergence, and a parameter-free reduced model bounds and ranks them in a prospective test. We show that the attainable gain grows only as $\kappa^2$, derive its coefficient in closed form, and thereby predict the threshold at which tail-risk optimization starts to pay. A controlled experiment that holds $\kappa$ fixed while varying the gate time and the noise correlation shows that this coefficient, not $\kappa$ alone, is what explains the gain. Under Ornstein–Uhlenbeck noise the comparison is unchanged across two decades of correlation time once that coefficient is held fixed, as the theory requires. Colored noise instead changes the shape of the tail, and we show that the tail-to-mean ratio measures the number of independent noise directions reaching the gate: one law that contains both the single-channel ratio 5.58 and the two-channel ratio 4.00, and that predicts each control's ratio from its geometry to 3%. Rotating the noise between two channels raises that number by a mechanism with no time structure, and the residuals from the two routes are statistically indistinguishable. Finally, adding trajectory-level $T_1$ relaxation gives a rare catastrophic channel whose effect on the tail-to-mean ratio we predict in closed form to 1% across a factor of 640 in its share of the mean loss — and which nonetheless leaves the selected pulse unchanged, because the gate requirement pins the channel to within a percent whatever the objective. Tail weight without control leverage buys nothing, a condition that $\kappa$ does not see.

## I. INTRODUCTION

Quantum gates are calibrated against parameters that drift: the qubit frequency wanders with two-level fluctuators and flux noise, and the drive amplitude with the electronics [1–3]. Robust optimal control designs pulses that tolerate such drifts, usually by maximizing the fidelity averaged over an ensemble of noise realizations [4–8] or by cancelling the leading orders of the error analytically or through filter functions [9–12]. The mean is a natural target, but it is not the quantity a fault-tolerant processor is most sensitive to. A surface-code threshold is a bound on the error of every gate in every round [13]; a gate that is excellent on average and poor on the few days the drift is large may be worse, in practice, than one that is slightly worse on average and never poor.

Risk measures from finance make this concern quantitative. The conditional value-at-risk,

$$\mathrm{CVaR}_\alpha(X) = \min_{\tau\in\mathbb{R}} \Big\{ \tau + \tfrac{1}{\alpha}\,\mathbb{E}\big[(X-\tau)_+\big] \Big\}, \qquad (1)$$

is the mean of the worst $\alpha$ fraction of outcomes [14, 15]; it is coherent [16], convex in the decision variables and straightforward to optimize by sampling. CVaR has been adopted as the objective of variational quantum optimization [17] and of quantum risk analysis [18], and risk-sensitive objectives based on exponential utility have been proposed for robust quantum gates [19], where they were observed to select controls different from those of the average fidelity. What has not been established is *when* a tail objective changes the answer. If the mean- and tail-optimal controls coincide, CVaR adds only estimation noise, since it is computed from a small fraction of the samples. If they differ, one needs to know by how much and for what physical reason.

In portfolio theory the analogous question has a classic answer: for normal or, more generally, elliptically distributed returns, every positively homogeneous translation-invariant risk measure produces the same efficient portfolios as the variance [14, 20]. The quantum problem differs in two ways. The loss is a periodic, bounded function of the error, not a linear one, and the noise can enter through several non-commuting generators and beyond linear order.

In this paper we give a sharp answer for single-qubit gates under quasi-static noise and test it at scale. Our

contributions are:

1. A theorem (Sec. II) stating that if the noise is Gaussian and couples linearly, every control's infidelity is a function of a single scalar, so the loss family is totally ordered by first-order stochastic dominance and all law-invariant monotone risk measures, including CVaR and exponential utility, share the mean-optimal control. The periodicity of $\sin^2(\phi/2)$ is handled by a monotone majorant with an explicit residual.

2. A characterization of how the ordering fails: two channels on non-parallel axes break it through a Schur-convexity effect, and second-order error does so through a pre-compensation mechanism, quantified by the ratio $\kappa$ of second- to first-order error and by a parameter-free reduced model. The tail gain is *quadratically* small in $\kappa$, with a closed form for its coefficient, which is why a sharp practical threshold exists at all.

3. A numerical test on a four-level transmon with an audited protocol that guards against optimization artefacts (Sec. III). The equivalence holds wherever the accumulated detuning phase variance satisfies $\sigma_\phi^2 \lesssim 0.05$, at three gate times and including the regime in which naive pulses leak strongly (Sec. IV A); the dominance premise itself holds over hundreds of candidate controls per setting (Sec. IV B).

4. An account of what CVaR does beyond the boundary: it deliberately miscalibrates the nominal gate, with an offset that matches a parameter-free prediction (Sec. IV E).

The practical result (Sec. V) is a test that can be run on the mean-optimal control alone and says whether a tail objective can help.

## II. THEORY

### A. Setup

Consider a qubit whose control-frame Hamiltonian contains a classical noise process $\eta(t)$ that couples linearly, so that the accumulated phase error of a control $u$ with toggling-frame modulation function $f_u(t)$ is

$$\phi[u] = \int_0^{T_g} f_u(t)\,\eta(t)\,dt. \tag{2}$$

The single-state infidelity of the resulting error rotation is $L = h(\phi) = \sin^2(\phi/2)$; for a coherent rotation the average-gate infidelity is $\frac{2}{3}h(\phi)$ [21], so all statements below hold for either figure of merit. We work under two hypotheses: **(A1)** $\eta$ is a Gaussian process; **(A2)** $\phi[u]$ is exactly the linear functional (2). Then $\phi[u] \sim \mathcal{N}(0, \sigma_u^2)$, and the whole loss distribution of a control depends on the control only through the scalar $\sigma_u$. A control cannot change the *shape* of the phase distribution, only its width.

### B. Equivalence

If $h$ were increasing in $|\phi|$, a narrower phase distribution would give a loss that is smaller at every quantile, and the theorem would be immediate. But $h$ is $2\pi$-periodic and decreases on $[\pi, 2\pi]$, which is exactly where CVaR looks. We therefore replace $h$ by its monotone majorant $\widetilde{L}(\phi) = h(\phi)\mathbf{1}(|\phi| \le \pi) + \mathbf{1}(|\phi| > \pi)$ [Fig. 1(a)] and bound the difference. Because $\widetilde{L} - h \le \mathbf{1}(|\phi| > \pi)$ and CVaR is monotone and subadditive, with $\mathrm{CVaR}_\alpha(\mathbf{1}_E) = \min(1, \Pr E/\alpha)$, we obtain the sandwich

$$\mathrm{CVaR}_\alpha\big(\widetilde{L}(\phi)\big) - r_\alpha(\sigma) \le \mathrm{CVaR}_\alpha\big(h(\phi)\big) \le \mathrm{CVaR}_\alpha\big(\widetilde{L}(\phi)\big), \tag{3}$$

with residual

$$r_\alpha(\sigma) = \frac{\Pr(|\phi| > \pi)}{\alpha} \le \frac{1}{\alpha}\sqrt{\frac{2}{\pi}}\,\frac{\sigma}{\pi}\,e^{-\pi^2/2\sigma^2}. \tag{4}$$

**Theorem 1** (Mean–CVaR near-equivalence)**.** *Under (A1)–(A2), let two controls have $\sigma_1 < \sigma_2$ and let $Z \sim \mathcal{N}(0,1)$. Then the mean loss $\mathbb{E}h(\phi) = \frac{1}{2}(1 - e^{-\sigma^2/2})$ is strictly larger for the second, and*

$$\mathrm{CVaR}_\alpha\big(h(\phi_2)\big) - \mathrm{CVaR}_\alpha\big(h(\phi_1)\big) \ge \Delta_{\mathrm{mono}} - r_\alpha(\sigma_2),$$

*where $\Delta_{\mathrm{mono}} = \mathrm{CVaR}_\alpha\big(\widetilde{L}(\sigma_2 Z)\big) - \mathrm{CVaR}_\alpha\big(\widetilde{L}(\sigma_1 Z)\big) \ge 0$. Hence the two objectives order the controls identically whenever $\Delta_{\mathrm{mono}} > r_\alpha(\sigma_2)$, and the mean-optimal control is CVaR-optimal whenever this holds against every competitor.*

*Proof sketch.* $\sigma_2|Z| \ge \sigma_1|Z|$ pointwise and $\widetilde{L}$ is nondecreasing in $|\phi|$, so $\widetilde{L}(\sigma_2 Z) \ge \widetilde{L}(\sigma_1 Z)$ pointwise and $\Delta_{\mathrm{mono}} \ge 0$ by monotonicity of CVaR. Apply the upper half of Eq. (3) to the first control and the lower half to the second. The mean follows from $\mathbb{E}\cos\phi = e^{-\sigma^2/2}$, which integrates over the periodicity exactly. Full proofs are in Sec. S1. □

The residual is negligible at any noise level of interest: at $\alpha = 0.05$ and $\sigma^2 = 0.05$ (mean infidelity 1.2%), $r_\alpha \approx 1.5 \times 10^{-43}$, while $\Delta_{\mathrm{mono}}$ between meaningfully different controls is of order $\sigma^2$. The theorem is a statement about the ranking of controls, not about the value of the objectives. The values differ by a control-independent factor:

**Proposition 1** (Tail amplification)**.** *As $\sigma^2 \to 0$, $\mathrm{CVaR}_\alpha(h(\phi))/\mathbb{E}h(\phi) \to 1 + 2z_\alpha\varphi(z_\alpha)/\alpha$, with $z_\alpha = \Phi^{-1}(1-\alpha/2)$ and $\varphi$ the standard normal density. The ratio is 3.25, 4.39, 5.58, 6.80 and 8.45 at $\alpha = 0.2$, 0.1, 0.05, 0.025 and 0.01.*

This follows from $h \approx \phi^2/4 = \sigma^2 Z^2/4$: the loss is a rescaled $\chi_1^2$ variable, and the tail average of $Z^2$ over $|Z| > z_\alpha$ is obtained by integration by parts. Proposition 1 gives a second, independent test of the theorem: in the equivalence regime *every* control must have the same tail-to-mean ratio.

The argument uses nothing specific to CVaR. It needs only that the family of loss distributions be totally ordered:

**Corollary 1** (Dominance principle)**.** *If $L_u \overset{d}{=} g(\tau(u)Y)$ for a fixed random variable $Y \geq 0$, a non-decreasing $g$ and a scalar $\tau(u) \geq 0$, then $\{L_u\}$ is totally ordered by first-order stochastic dominance, and every risk measure that is monotone under that order (the mean, every $\mathrm{CVaR}_\alpha$, the exponential utility of Ref. [19], every law-invariant coherent measure [22, 23]) has the same set of minimizers.*

Two extensions follow immediately. For Ornstein–Uhlenbeck noise, or any stationary Gaussian noise with memory, $\phi[u]$ in Eq. (2) is still Gaussian: memory changes the size of $\sigma_u^2$, not the equivalence. Likewise, several jointly Gaussian noise sources that couple into the *same* generator (for example several sources of frequency noise) give a Gaussian $\phi[u]$ whose variance is a quadratic form in their coupling functions (Sec. S1.A).

### C. How the ordering fails

*Non-parallel channels.* Detuning noise rotates the qubit about $z$ and amplitude noise about the drive axis. To leading order the per-draw error is then a rotation vector, and after diagonalization in the noise metric the loss becomes $L \approx L_0 + \lambda_1(u)Z_1^2 + \lambda_2(u)Z_2^2$ with independent standard normals $Z_i$: a control has two weights instead of one scale.

**Proposition 2** (Two non-parallel channels)**.** *Let $L_u = \lambda_1 Z_1^2 + \lambda_2 Z_2^2$ with $\lambda_i \geq 0$. (i) $\mathbb{E}L_u = \lambda_1 + \lambda_2$. (ii) For every $\alpha$, $\mathrm{CVaR}_\alpha(L_u)$ is Schur-convex in $(\lambda_1, \lambda_2)$: at fixed mean it grows as the split becomes more unequal, from $1 + \ln(1/\alpha)$ (equal weights; 4.00 at $\alpha = 0.05$) to $1 + 2z_\alpha\varphi(z_\alpha)/\alpha$ (one axis; 5.58), in units of the mean. (iii) Two controls with equal mean and different splits are not ordered by first-order dominance. Componentwise-ordered sorted weights are sufficient for dominance; a common ratio $\lambda_2/\lambda_1$ is therefore sufficient for a totally ordered family but not necessary.*

*Proof sketch.* In Eq. (1), for fixed $\tau$ the objective $\tau + \alpha^{-1}\mathbb{E}(\lambda_1 Z_1^2 + \lambda_2 Z_2^2 - \tau)_+$ is convex and symmetric in $\lambda$, hence Schur-convex [24], and the property survives the minimum over $\tau$. For (iii), dominance together with equal means forces equality in law, but the variances $2(\lambda_1^2 + \lambda_2^2)$ differ. □

A sum of two independent squared Gaussians has a lighter tail than one squared Gaussian of the same mean, so a control that spreads its error over two axes has a better tail at the same mean [Fig. 1(c)]. This is a genuine crossing of distributions, and the simplest situation in which mean and CVaR can disagree even with exactly linear coupling.

*The effective dimension.* Propositions 1 and 2 are the first two cases of one law. Let the loss be a positive quadratic form in the noise, $L = \sum_i w_i Z_i^2$ with $w_i \geq 0$ and $Z_i$ independent standard normals — the general shape of a first-order error — and define the *effective dimension* as the participation ratio of the weights,

$$m_{\mathrm{eff}} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2} \in \left[1,\ \#\{w_i > 0\}\right]. \tag{5}$$

**Proposition 3** (Effective dimension)**.** *Let $\chi^2_{m,1-\alpha}$ denote the $(1-\alpha)$-quantile of $\chi^2_m$ and*

$$\rho_\alpha(m) = \frac{1}{\alpha}\, P\big(\chi^2_{m+2} > \chi^2_{m,1-\alpha}\big). \tag{6}$$

*Then (i) for $m$ equal weights $\mathrm{CVaR}_\alpha(L)/\mathbb{E}L = \rho_\alpha(m)$ exactly; (ii) $\rho_\alpha(1) = r_\alpha$ and $\rho_\alpha(2) = 1 + \ln(1/\alpha)$, the two endpoints of Proposition 2; (iii) $\rho_\alpha$ is strictly decreasing in $m$, with $\rho_\alpha(m) \to 1$; and (iv) for unequal weights $\rho_\alpha(m_{\mathrm{eff}})$ estimates the ratio, from below in 99% of cases and within 6.5% at $\alpha = 0.05$ over 400 random spectra of 2–10 directions (Sec. S1.C).*

*Proof sketch.* For equal weights $L \propto \chi^2_m$, and $\mathbb{E}[\chi^2_m \mathbf{1}\{\chi^2_m > q\}] = m\,P(\chi^2_{m+2} > q)$ by the standard size-biasing identity, which gives (i) and, on substituting $m = 1$ and $m = 2$, (ii); $\chi^2_2/2$ is exponential, whence the logarithm. Monotonicity in (iii) is the concentration of $\chi^2_m/m$. Part (iv) replaces the spectrum by the equal-weight spectrum of the same mean and the same second moment; it is numerical, and the error grows as $\alpha \to 0$, where the tail is governed by the largest weight rather than by $m_{\mathrm{eff}}$. □

Read in the other direction, Proposition 3 makes the tail-to-mean ratio a *measurement* of how many independent noise directions reach the gate. A control whose error is carried by a single direction sits at $r_\alpha$; spreading it over more directions lowers the ratio along a known curve, and inverting $\rho_\alpha$ recovers $m_{\mathrm{eff}}$ from one loss histogram, with no optimization and no model of the device. This is the quantity Proposition 2 says $\mathrm{CVaR}_\alpha$ is Schur-convex in, in a form that can be read off an experiment. Section IV I measures it across two decades of noise correlation time and Section IV J along an independent route.

*A rare catastrophic channel.* Both propositions above describe a loss whose randomness is continuous. A relaxation event is not: it fires with small probability and, when it does, costs an amount that does not scale with the noise. Such a channel changes the ratio in a way that depends on one number only.

**Proposition 4** (Rare-channel mixture)**.** *Let $L = L_c + \ell_j \mathbf{1}_J$, where the continuous part obeys $\mathrm{CVaR}_\alpha(L_c) =$*

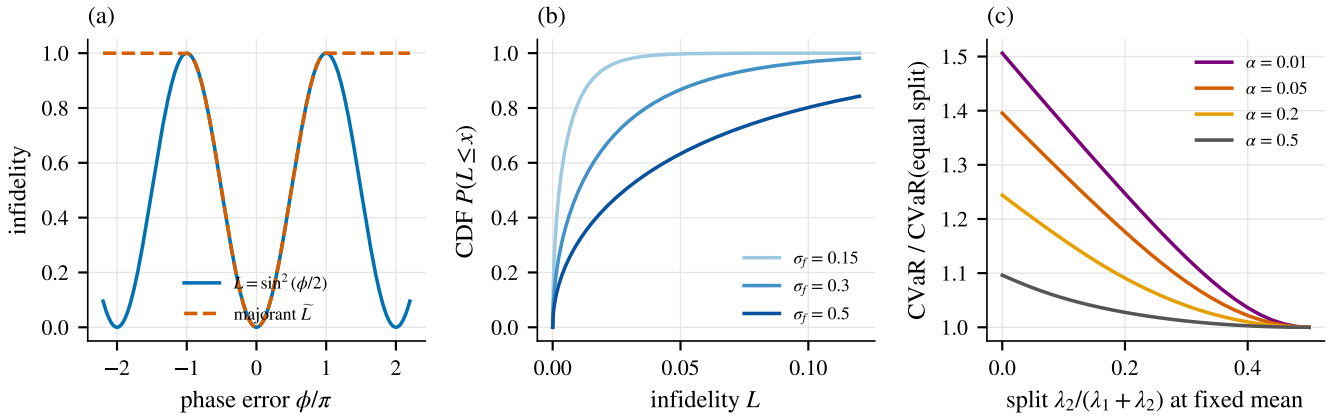



FIG. 1. The three ingredients of the theory. (a) The infidelity $\sin^2(\phi/2)$ is periodic and hence not monotone in $|\phi|$; its majorant $\widetilde{L}$ agrees with it for $|\phi| \le \pi$ and is monotone. (b) Cumulative distributions of the infidelity for three phase widths: the curves never cross, so a narrower phase distribution is better at every quantile (Theorem 1). (c) With two error axes the ordering fails. At a fixed mean $\lambda_1 + \lambda_2$, CVaR depends on how the error is split between the axes; it is lowest for an equal split and highest when one axis carries all the error, by 40% at $\alpha = 0.05$ and 51% at $\alpha = 0.01$ (Proposition 2).

$\rho_\alpha(m_{\text{eff}})\,\mathbb{E}L_c$, *the event* $J$ *is independent of* $L_c$ *with* $P(J) = p < \alpha$, *and* $\ell_j$ *exceeds the* $(1-\alpha)$*-quantile of* $L_c$. *Write the* jump share

$$\beta = \frac{p\,\ell_j}{\mathbb{E}L_c}, \tag{7}$$

*so that* $\mathbb{E}L = (1+\beta)\,\mathbb{E}L_c$. *Then, to first order in* $p/\alpha$,

$$\frac{\mathrm{CVaR}_\alpha(L)}{\mathbb{E}L} = \frac{\rho_\alpha(m_{\text{eff}}) + \beta/\alpha}{1+\beta}. \tag{8}$$

*The ratio increases monotonically from* $\rho_\alpha(m_{\text{eff}})$ *at* $\beta \to 0$ *to the hard ceiling* $1/\alpha$ *at* $\beta \to \infty$, *and exceeds* $\rho_\alpha$ *for every* $\beta > 0$ *whenever* $\rho_\alpha < 1/\alpha$.

*Proof sketch.* Because $\ell_j$ sits above the $(1-\alpha)$-quantile and $p < \alpha$, the worst $\alpha$ of the mass is the whole of $J$ together with the worst $\alpha - p$ of $J^c$, so $\alpha\mathrm{CVaR}_\alpha(L) = p\,\ell_j + (\alpha - p)\mathrm{CVaR}_{\alpha-p}(L_c) + O(p\,\mathbb{E}L_c)$. Letting $p/\alpha \to 0$ gives $\mathrm{CVaR}_\alpha(L) = \beta\,\mathbb{E}L_c/\alpha + \rho_\alpha\,\mathbb{E}L_c$, and dividing by $\mathbb{E}L$ gives Eq. (8); $p/\alpha \le 0.04$ throughout Section IV K. Full statement and error term in Sec. S1.D. □

Equation (8) is a statement about how a rare channel reshapes the loss *distribution*, and in that role it is exact to the stated order. It is worth flagging at once what it does not say. It carries no information about whether the rare branch can be *moved* by the control, and in Sec. IV M we show that the suppression of the tail gain is governed by exactly that—by how much of the tail loss the pulse can reduce, for which the bulk share is an exact proxy only when the rare channel is control-inert. Relaxation is that limit; a fluctuator is not.

Equation (8) has no free parameters: $\beta$, $p$ and $\ell_j$ are each measurable on a given control. It also makes the reweighting explicit. A tail objective weights the jump branch by $1/\alpha$ and the bulk by $\rho_\alpha$, so relative to the mean it amplifies the rare channel by

$$\Xi_\alpha = \frac{1}{\alpha\,\rho_\alpha(m_{\text{eff}})}, \tag{9}$$

which is 3.58 at $\alpha = 0.05$, $m_{\text{eff}} = 1$. Whether that amplification moves the optimizer is a separate question; Section IV K answers it in the negative.

*Nonlinear coupling.* The second route violates (A2). Expand the error rotation vector of a control in whitened noise coordinates $\mathbf{z}$ (independent standard normals),

$$\boldsymbol{\epsilon}(\mathbf{z}) = \mathbf{e}_0 + A\mathbf{z} + Q(\mathbf{z}) + O(|\mathbf{z}|^3), \tag{10}$$

with $\mathbf{e}_0$ the coherent (noise-free) error, $A$ the linear response and $Q$ a vector-valued quadratic form. We define the *second-order ratio*

$$\kappa = \|Q\|_F / \|A\|_F, \tag{11}$$

so that (A2) is $\kappa = 0$. On a driven qubit, a detuning $\delta$ does not commute with the drive; the error is a linear phase only to first order in $\delta/\Omega$, with Rabi rate $\Omega \sim \pi/T_g$. Since $\delta T_g$ is the accumulated phase, (A2) holds approximately only while $\sigma_\phi^2 = (\sigma_\delta T_g)^2$ is small, and $\kappa$ grows with $\sigma_\phi$. The quadratic term is qualitatively different from the linear one: for a single noise variable it is $\mathbf{q}z^2$, which points the same way for every draw and grows fastest in exactly the rare draws that form the tail. A coherent offset $\mathbf{e}_0$ can partly cancel it, and how large the best offset is depends on whether one cancels it for typical draws or for the worst ones.

**Proposition 5** (Reduced second-order model)**.** *Fix* $A$ *and* $Q$, *let the loss be* $L = \boldsymbol{\epsilon}^{\mathsf{T}} W \boldsymbol{\epsilon}$ *with* $W \succ 0$, *and let each objective choose only* $\mathbf{e}_0$. *Denote by* $R_\alpha^{\text{model}}$ *the resulting relative CVaR gain of the CVaR-optimal*

*offset over the mean-optimal one. (i) The mean optimum is* $\mathbf{e}_0 = -\mathbb{E}\,Q(\mathbf{z})$. *(ii)* $R_\alpha^{\text{model}}$ *depends on the noise strength only through* $\kappa$ *and on the normalized geometry* $(A/\|A\|, Q/\|Q\|)$. *(iii)* $R_\alpha^{\text{model}} = 0$ *at* $\kappa = 0$.

*Proof sketch.* (i) $\mathbb{E}L$ is quadratic in $\mathbf{e}_0$. (ii) Factor $\|A\|$ out of $\boldsymbol{\epsilon}$; the factor cancels in the ratio. (iii) At $\kappa = 0$ the noise part is symmetric under $\mathbf{z} \to -\mathbf{z}$, so $\mathrm{CVaR}_\alpha(L(\mathbf{e}_0))$ is even and convex in $\mathbf{e}_0$ and is minimized at $\mathbf{e}_0 = 0$, the mean optimum. □

**Remark 1** (Pre-compensation)**.** In the one-dimensional limit in which the linear error is cancelled, $\boldsymbol{\epsilon} \approx \mathbf{e}_0 + \mathbf{q}z^2$. Writing $\mathbf{e}_0 = -c\,\mathbf{q}$, the mean loss $\propto \mathbb{E}(z^2 - c)^2$ is minimized at $c = \mathbb{E}z^2 = 1$, an offset that cancels the second-order error of a typical draw. $\mathrm{CVaR}_\alpha$ is minimized at $c = 2.76$ for $\alpha = 0.05$ and $c = 1.70$ for $\alpha = 0.2$, where the offset cancels the error of the tail draws instead. The CVaR-optimal gate is deliberately miscalibrated.

**Proposition 6** (Quadratic smallness of the tail gain)**.** *In the setting of Proposition 5, let* $L_0 = \mathbf{z}^\mathsf{T} A^\mathsf{T} W A \mathbf{z}$ *have a continuous positive density* $\varrho$ *at its* $\alpha$*-quantile* $\tau$*, and write the offset as* $\mathbf{e}_0 = \kappa\mathbf{f}$. *Then*

$$\mathrm{CVaR}_\alpha(L) = \mathrm{CVaR}_\alpha(L_0) + \kappa^2 J(\mathbf{f}) + O(\kappa^3), \qquad (12)$$

*where, with* $Q = Q(\mathbf{z})$ *and* $\mathbf{1}_\tau = \mathbf{1}\{L_0 > \tau\}$,

$$J(\mathbf{f}) = \tfrac{1}{\alpha}\mathbb{E}\big[(Q+\mathbf{f})^\mathsf{T} W (Q+\mathbf{f})\,\mathbf{1}_\tau\big] + \tfrac{\varrho(\tau)}{2\alpha}\mathrm{Var}\big[2(A\mathbf{z})^\mathsf{T} W (Q+\mathbf{f}) \,\big|\, L_0 = \tau\big]$$

*is a positive-definite quadratic form in* $\mathbf{f}$. *Consequently*

$$R_\alpha(\kappa) = C\,\kappa^2 + O(\kappa^3), \qquad C = \frac{J(\mathbf{f}_{\text{mean}}) - \min_{\mathbf{f}} J}{\mathrm{CVaR}_\alpha(L_0)}, \qquad (13)$$

*with* $\mathbf{f}_{\text{mean}} = -\mathbb{E}\,Q(\mathbf{z})$. *If moreover the second-order error is parallel to the first-order one and a single noise variable dominates, then*

$$c_\alpha^\star = 1 + z_\alpha^2\Big(1 - \tfrac{1}{r_\alpha}\Big), \qquad C_\alpha = (c_\alpha^\star - 1)^2, \qquad (14)$$

*where* $r_\alpha = 1 + 2z_\alpha\varphi(z_\alpha)/\alpha$ *is the tail-amplification ratio of Proposition 1. Equation* (14) *is the* $m_{\text{eff}} = 1$ *case; at larger effective dimension we have no closed form and evaluate* $J$ *numerically, which is what the data below are compared against.*

*Proof sketch.* At $\kappa = 0$ the noise enters only through $A\mathbf{z}$, which is symmetric under $\mathbf{z} \to -\mathbf{z}$, so the $\alpha$-tail of $L_0$ is a symmetric two-sided set. A coherent offset improves the loss on one side of that tail by exactly what it costs on the other, and the $O(\kappa)$ terms cancel; this cancellation is the reason a threshold exists at all. Expanding $L = L_0 + \kappa h_\mathbf{f} + \kappa^2 g_\mathbf{f}$ with $h_\mathbf{f} = 2(A\mathbf{z})^\mathsf{T} W(Q+\mathbf{f})$ odd and $g_\mathbf{f} = (Q+\mathbf{f})^\mathsf{T} W (Q+\mathbf{f})$ even, the envelope theorem applied to Eq. (1) gives $\partial_\kappa \mathrm{CVaR} = \alpha^{-1}\mathbb{E}[(h + 2\kappa g)\mathbf{1}_\tau]$, which vanishes at $\kappa = 0$ by that symmetry, and one further differentiation, using $\partial_\kappa \mathrm{VaR} = \mathbb{E}[h \mid L_0 = \tau]$, gives Eq. (12). In the scalar case the level set $\{L_0 = \tau\}$ is the two points $\pm z_\alpha$, the conditional variance is explicit, and minimizing the resulting quadratic gives Eq. (14). Details in Sec. S1.G. □

**Corollary 2** (Threshold)**.** *A tail objective can beat the mean-optimal control by a relative margin* $R_{\min}$ *only if* $\kappa \geq \sqrt{R_{\min}/C}$. *For one dominant direction and* $\alpha = 0.05$ *this reads* $\kappa \geq 0.32\sqrt{R_{\min}}$.

Equation (13) turns the empirical threshold of this paper into a statement with a constant. It also predicts that the optimal offset is not a single number: $c^\star$ falls from 4.15 at $\kappa \to 0$ to the 2.76 of Remark 1 as $\kappa \to \infty$ (at $\alpha = 0.05$), and the gain saturates at the value of that limit rather than growing as $\kappa^2$ forever.

The value $c = 2.76$ follows from the first-order condition $\mathbb{E}[z^2 - c \mid \text{tail}] = 0$; at the optimum the tail consists of the largest draws together with the draws near $z = 0$, which the offset now overshoots (Sec. S1.F). Proposition 5 makes a quantitative, parameter-free prediction for any control once its $A$ and $Q$ are measured; Remark 1 predicts the sign and size of the offset a CVaR optimizer should find. Both are tested in Sec. IV E.

## III. MODEL AND PROTOCOL

*Transmon and noise.* We simulate an $N$-level transmon ($N = 4$, or $N = 2$ as a leakage-free control) in the frame rotating at the qubit frequency,

$$H(t;\delta,\gamma) = \tfrac{\alpha_{\text{anh}}}{2} a^\dagger a^\dagger a a + \delta\, a^\dagger a + (1+\gamma)\big[u_I(t)\tfrac{a+a^\dagger}{2} + u_Q(t)\tfrac{i(a^\dagger - a)}{2}\big], \qquad (15)$$

with $\alpha_{\text{anh}}/2\pi = -200$ MHz, $T_1 = 45.5$ $\mu$s, $T_2^* = 16.8$ $\mu$s and a 50 mK thermal background [2, 25], and with dissipation entering through the Lindblad equation [26]. The detuning $\delta$ and the relative amplitude error $\gamma$ are quasistatic, drawn once per gate from a bivariate normal distribution with widths $\sigma_\delta$, $\sigma_\gamma$ and correlation $\rho$. The target is $X_{\pi/2}$ on the computational subspace, and the loss of a draw is $L = 1 - \bar{F} + 7P_{\text{leak}}$, with $\bar{F}$ averaged over four input states and $P_{\text{leak}}$ the population above level 1.

TABLE I. The coefficients of Proposition 6 for one dominant noise direction, i.e. $m_{\text{eff}} = 1$: the tail-to-mean ratio, the ratio at two equal directions for comparison, the optimal offset $c_\alpha^\star$ in units of $\kappa|\mathbf{q}|$, the gain coefficient $C_\alpha$, and the offset in the opposite limit $\kappa \to \infty$ (Remark 1).

| $\alpha$ | 0.20 | 0.10 | 0.05 | 0.025 | 0.01 |
|---|---|---|---|---|---|
| $r_\alpha = \rho_\alpha(1)$ (Prop. 1) | 3.25 | 4.39 | 5.58 | 6.80 | 8.45 |
| $\rho_\alpha(2)$ (Prop. 3) | 2.61 | 3.30 | 4.00 | 4.69 | 5.61 |
| $c_\alpha^\star$ ($\kappa \to 0$) | 2.14 | 3.09 | 4.15 | 5.29 | 6.85 |
| $C_\alpha$ | 1.29 | 4.37 | 9.94 | 18.4 | 34.2 |
| $c$ at $\kappa \to \infty$ | 1.70 | – | 2.76 | – | – |

*Controls.* Each quadrature is a sum of Hann-windowed sine modes (17 per quadrature at $T_g$ = 12.7 ns), bounded smoothly by $u = A\tanh(\tilde{u}/A)$ with $A = 2.5\pi/T_g$, so that the amplitude budget is equally tight at every gate time, and passed through a fourth-order, 300 MHz zero-phase Butterworth filter. The optimized controls use the full budget.

*Objectives and simulators.* The mean and $\mathrm{CVaR}_\alpha$ objectives ($\alpha = 0.01$–$0.2$) are computed on the same $K = 128$ training draws, with CVaR in the exact Rockafellar–Uryasev form (at $\alpha = 0.05$, six draws at weight 0.156 and one at 0.0625), and minimized by L-BFGS-B [27] with identical tolerances. Optimization uses a closed-system simulator with exact gradients from the Daleckii–Krein formula [28]. Because $T_1$ and $T_2^*$ add a nearly constant loss to every quasi-static draw, and $\mathrm{CVaR}_\alpha(L + c) = \mathrm{CVaR}_\alpha(L) + c$, this does not move the optimum. *Every reported number*, however, is a re-evaluation of the saved controls in the full Lindblad model on a held-out test ensemble of 1000 draws; a validation ensemble of 300 draws is used only to select between restarts. The two simulators agree on the absolute tail gap to a median ratio of 1.000 (Sec. S5).

*Fairness protocol.* A small difference between two optimized controls is easily produced by the optimization rather than by the objectives. Every point is therefore run under the following rules (Sec. S3.A). Both objectives receive the same number of restarts (four, or twenty in a strong-search check). One CVaR restart is warm-started from the selected mean control $u_{\mathrm{mean}}$; since L-BFGS-B accepts only descent steps, it cannot end with a higher training CVaR, and this *in-sample audit* is checked at every point. The mean objective receives one extra continuation run, so that CVaR never wins by being optimized longer. For each CVaR control $u_{\mathrm{CVaR}}$, the mean optimizer is restarted from $u_{\mathrm{CVaR}}$ (*reverse warm start*); if it walks back to the loss of $u_{\mathrm{mean}}$, the two optima share a basin, and if it finds a better mean control, the multistart had missed a basin. Finally, $u_{\mathrm{CVaR}}$ is compared with the best mean control found by any route (*conservative comparator*), so that a positive result is a lower bound on the CVaR advantage. Cells in which the comparator was upgraded by a reverse start are marked, and none is dropped.

*Statistics.* The outcome is the relative tail gain

$$R_{\mathrm{tail}}(\alpha) = 1 - \frac{\mathrm{CVaR}_\alpha(L[u_{\mathrm{CVaR}}])}{\mathrm{CVaR}_\alpha(L[u_{\mathrm{mean}}])}, \tag{16}$$

so that $R_{\mathrm{tail}} = 0.2$ means a 20% lower worst-$\alpha$ error. Confidence intervals are from a paired bootstrap over the test draws [29] (2000 resamples, the same resampled indices for both controls). Replicating points with independent noise seeds shows that out-of-sample $R_{\mathrm{tail}}$ scatters by about $\pm 0.04$ where the objectives are equivalent, and that $\alpha = 0.01$ (1.3 training draws) overfits; we therefore report $\alpha = 0.05$ and 0.2 (Sec. S3.C). Because the $T_1/T_2$ floor enters the denominator of Eq. (16), full-Lindblad values of $R_{\mathrm{tail}}$ are diluted relative to closed-system ones, and therefore conservative (Sec. S5).

*Diagnostics.* For each saved control we extract the error rotation vector of every test draw from the qubit block of the closed-system propagator and fit Eq. (10) by least squares in whitened noise coordinates, which gives $\mathbf{e}_0$, $A$, $Q$ and $\kappa$. From the fitted $A$ and $Q$ of the mean control, $R_{0.05}^{\mathrm{model}}$ is computed with no free parameter from $6 \times 10^4$ samples, using the four-state loss metric $W = \mathrm{diag}(3, 3, 2)/16$, which reproduces the simulated loss to leading order (Sec. S2.E).

## IV. RESULTS

Unless stated otherwise, all values are full-Lindblad test-ensemble values at $\alpha = 0.05$ against the best mean control found. The experiments and all per-point results are listed in Sec. S6 and Sec. S13.

### A. The equivalence boundary

The setting closest to Theorem 1 is a two-level qubit with detuning noise alone. We scanned $\sigma_\delta$ at three gate times, $T_g$ = 8, 12.7 and 20 ns, choosing $\sigma_\delta$ so that the accumulated phase variance $\sigma_\phi^2 = (\sigma_\delta T_g)^2$ takes the same six values at every gate time (Table II, Fig. 2).

For $\sigma_\phi^2 \le 0.04$, $R_{\mathrm{tail}}$ is zero within its confidence interval at all three gate times: the numerical optimizers return controls with the same tail, as the theorem requires. Beyond $\sigma_\phi^2 \approx 0.1$ the gain grows smoothly and monotonically, to 0.19–0.29 at $\sigma_\phi^2 = 1.6$. The curves collapse: at $\sigma_\phi^2 = 0.16$ the three gate times give 0.018, 0.029 and 0.021, and at 0.41 they give 0.051, 0.077 and 0.056. A 2.5-fold change of gate time at fixed $\sigma_\phi$ thus leaves both the onset and the size of the effect nearly unchanged, as the $\delta/\Omega$ argument of Sec. II C predicts; only the shortest gate falls somewhat lower at the largest phase variance.

The divergence is reproducible and is not an optimization artefact. At $\sigma_\phi^2 = 1.6$ three independent noise seeds give $R_{\mathrm{tail}} = 0.275$, 0.271 and 0.314, and a second, independent set of runs with eight restarts gives 0.274, 0.276 and 0.290 (Sec. IV B). At every divergent point the reverse warm start returns to $u_{\mathrm{mean}}$: the two optima

TABLE II. Relative tail gain $R_{\mathrm{tail}}(0.05)$ on a two-level qubit with detuning noise only, at matched accumulated phase variance, full Lindblad, against the best mean control. At $T_g = 12.7$ ns the 95% intervals are $[-0.003, 0.007]$ at $\sigma_\phi^2 = 0.04$ and $[0.026, 0.032]$, $[0.068, 0.085]$, $[0.247, 0.298]$ at the three largest values. °Comparator upgraded by a reverse start.

| $\sigma_\phi^2$ | $\approx 0.005$ | 0.04 | 0.16 | 0.41 | 1.6 |
|---|---|---|---|---|---|
| $T_g = 8$ ns | $-0.002$° | $+0.001$ | $+0.018$ | $+0.051$ | $+0.187$ |
| $T_g = 12.7$ ns | $-0.002$ | $+0.002$ | $+0.029$ | $+0.077$ | $+0.275$ |
| $T_g = 20$ ns | $-0.001$° | $+0.003$ | $+0.021$ | $+0.056$° | $+0.292$° |

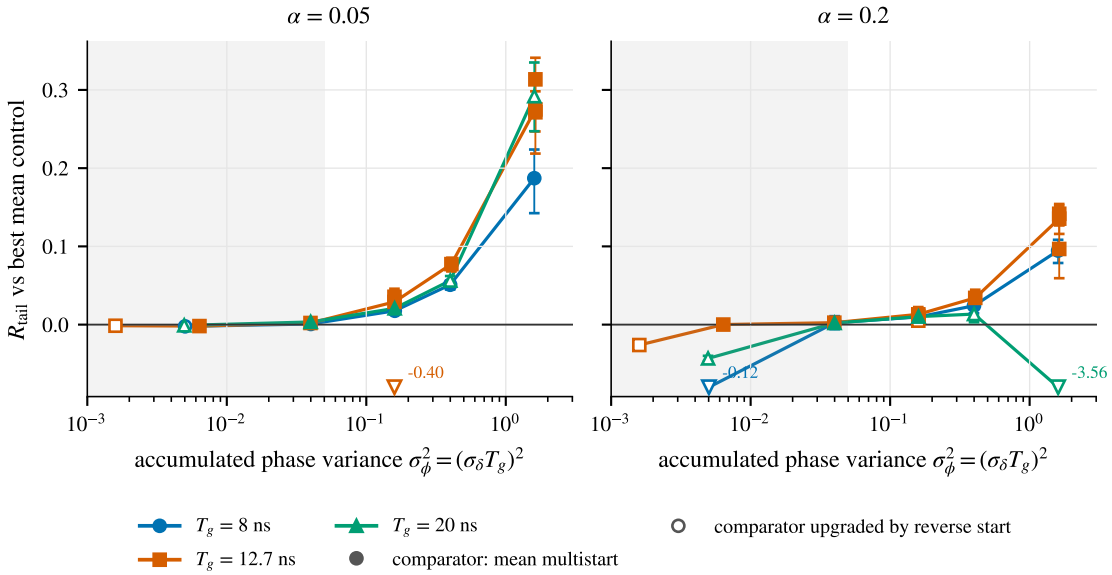



FIG. 2. The equivalence boundary. $R_{\text{tail}}$ versus accumulated phase variance $\sigma_\phi^2 = (\sigma_\delta T_g)^2$ at three gate times, two-level qubit, detuning noise only, against the best mean control. Shaded: $\sigma_\phi^2 < 0.05$, where the residual of Theorem 1 is negligible. Open markers: comparator upgraded by a reverse start; unconnected points at $T_g = 12.7$ ns are replication seeds; values below the axis are marked at its lower edge (comparator asymmetry, Sec. S3.D).

lie in one basin, and CVaR shifts the solution within it rather than selecting a different pulse family. The shift is small, a few percent in the norm of the pulse envelopes (Sec. S7 E 0 c).

### B. A direct test of the dominance premise

The comparison above tests a consequence of the theorem, namely that two optimizers return the same control. The theorem states more: under (A1)–(A2) the loss distributions of *all* controls are rescaled copies of one shape, so all pairs of controls are ranked identically by all risk measures. We tested this directly over large pools of controls.

At each of 42 settings (two levels at $T_g = 8$, 12.7 and 20 ns plus two additional noise seeds at 12.7 ns; four levels at 12.7 and 20.8 ns; the six phase variances above), the full comparison was repeated with eight restarts per objective and every L-BFGS-B iterate of every run was recorded. A thinned subset of iterates together with every final control forms a pool of 376–491 candidate controls, which we evaluated on a fresh ensemble of 4000 draws split into two independent halves. Over the *competitive* candidates (mean loss within a factor of two of the best) we computed: the Kendall rank correlation $\tau$ between the orderings by mean and by $\text{CVaR}_{0.05}$ (theorem: $\tau = 1$); the fraction of *confirmed rank reversals*, among pairs separated by more than 2% in both measures on both halves, in which the two measures disagree with the same orientation on both halves (theorem: 0); the fraction of pairs with at least 10% difference in mean whose quantile functions cross by more than 2% in the upper half of the distribution (theorem: 0); the ratio $\text{CVaR}_\alpha$/mean of every candidate (Proposition 1: 5.58 and 3.25); and a cross-fitted *pool regret* $R_{\text{pool}}$, the CVaR gain of the best-CVaR over the best-mean candidate when both are selected on one half and scored on the other. The statistics were validated before the run on synthetic families: an exactly ordered family $\sin^2(\sigma_i Z/2)$ gives $\tau = 1$ and no reversals, and the pre-compensation family $(Z^2 - c_i)^2$ of Remark 1 gives $\tau = -0.46$ and 76% reversals.

On the two-level qubit the premise holds exactly where the theorem says it should (Fig. 3). At $\sigma_\phi^2 \le 0.0064$ there is not one confirmed rank reversal among 19 639 well-separated pairs, over ten settings, three gate times and three noise seeds. At $\sigma_\phi^2 = 0.04$ reversals stay below 0.8%, and no pair crosses in the upper half of the distribution at any two-level setting with $\sigma_\phi^2 \le 0.04$. The shape prediction holds to 2%: the median $\text{CVaR}_{0.05}$/mean is 5.49–5.60 against 5.58, and $\text{CVaR}_{0.2}$/mean is 3.23–3.24 against 3.25, with an interquartile spread below 1% across candidates. The normalized loss distributions of all competitive candidates fall on the single $\chi_1^2$ curve [Fig. 4(b)]. Even with a free choice among hundreds of controls, CVaR has nothing to gain: $R_{\text{pool}}$ lies between $-0.002$ and $+0.001$. The rank correlation is high but below one ($\tau = 0.90$–$0.98$) because near-identical late iterates are ordered by differences below the sampling noise; the confirmed-reversal statistic, which requires a clear separation, is the sharper test.

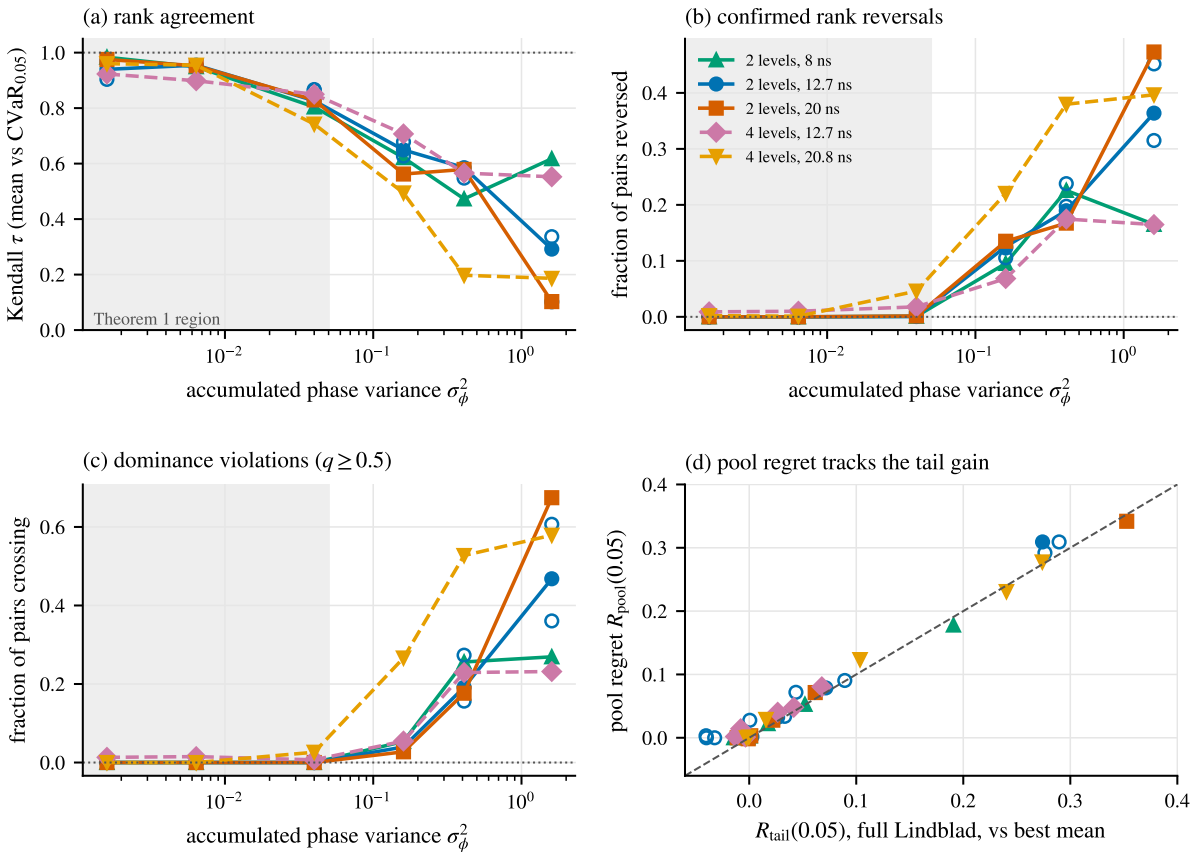



FIG. 3. Direct test of the dominance premise over pools of 376–491 candidate controls per setting (closed markers: first noise seed; open: two replication seeds). Shaded: $\sigma_\phi^2 < 0.05$. Dotted: the theorem's prediction. (a) Kendall $\tau$ between the rankings by mean and by $\mathrm{CVaR}_{0.05}$. (b) Fraction of well-separated pairs whose rankings are reversed, confirmed on two independent halves of a fresh 4000-draw ensemble. (c) Fraction of pairs whose quantile functions cross in the upper half of the distribution. (d) Cross-fitted pool regret against the full-Lindblad $R_{\mathrm{tail}}$ of the standard comparison (Spearman 0.90, 42 settings); dashed: equality.

As $\sigma_\phi^2$ grows the premise fails, at the same threshold as the optimized comparison. Reversals reach 10–13% of pairs at $\sigma_\phi^2 = 0.16$, 17–24% at 0.41 and 17–47% at 1.6; upper-half crossings follow (3–5%, 16–27% and 27–68%), and $\tau$ falls to 0.10–0.62. The onset is the same at all gate times and seeds. The shapes stop collapsing: the lighter-tailed candidates carry more loss in the bulk, the signature of pre-compensation (Sec. IV E). The pool regret tracks the optimized-pair $R_{\mathrm{tail}}$ (Spearman 0.90, $p = 4 \times 10^{-16}$), so the optimized pair captures what the whole family of controls can offer.

One analysis choice needs stating. As first specified, the crossing test compared entire quantile functions without a mean separation, and reported 4–31% crossing pairs at $\sigma_\phi^2 \leq 0.04$, where every other statistic shows exact equivalence. All of these crossings lie in the lower half of the distribution (median quantile 0.05–0.19, none above 0.46), where a $\chi_1^2$ loss is tiny and is set by the residual coherent error rather than by the noise; in 84–98% of the crossing pairs the control with the lower mean has the larger coherent error (29–78% in one seed). Theorem 1 assumes no coherent error, and the mean and CVaR are both governed by the upper half. We therefore restricted the test to $q \geq 0.5$ and to pairs that differ in mean, applied the restriction uniformly, and report the unrestricted values in Sec. S8.

### C. Leakage does not break the equivalence

The four-level transmon adds a physically distinct violation of (A2): population transfer to levels $\geq 2$ is a strongly nonlinear function of the calibration errors. The nominal leakage of a Hann pulse of the correct area falls from 3% at 6.3 ns to the thermal floor ($6.8 \times 10^{-5}$) by about 20 ns, exceeding five times the floor below $T_g = 12.7$ ns (Sec. S7.A).

With both noise channels at their nominal widths ($\sigma_\delta/2\pi = 2.5$ MHz, $\sigma_\gamma = 2\%$), the mean and CVaR controls coincide within $|R_{\mathrm{tail}}| \leq 0.026$ throughout the leakage-dominated region, $6.3 \leq T_g \leq 12.7$ ns, with four levels and with two (Fig. 5). Leakage is not ab-

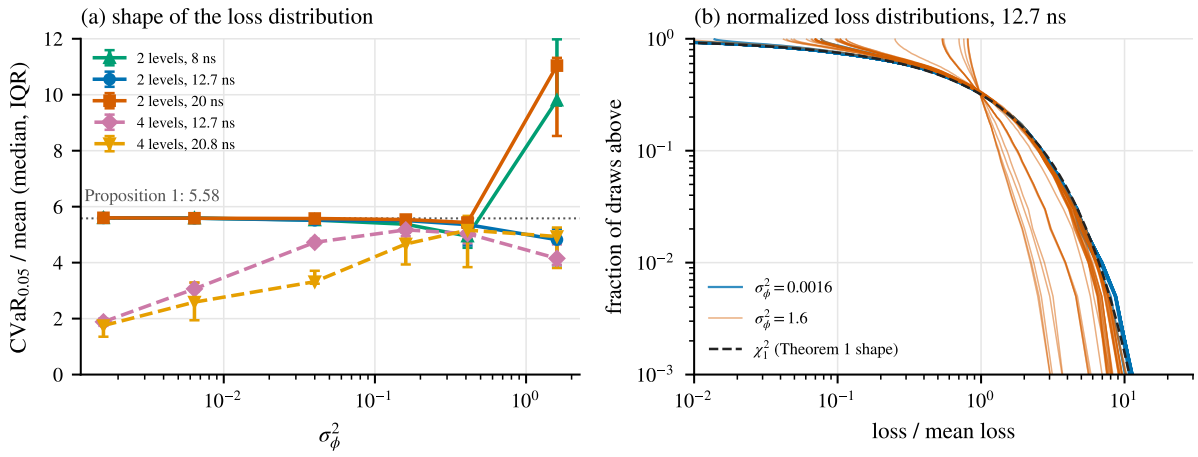



FIG. 4. Shape test. (a) Median $\mathrm{CVaR}_{0.05}$/mean over the competitive candidates (bars: interquartile range) versus $\sigma_\phi^2$. Dotted: 5.58 (Proposition 1). On the transmon the ratio is lower at small noise because a noise-independent leakage floor makes up 56–84% of the mean loss. (b) Complementary distributions of the loss divided by the mean loss for 40 competitive candidates at $T_g = 12.7$ ns. At $\sigma_\phi^2 = 0.0016$ (blue) all collapse onto the $\chi_1^2$ shape (dashed); at $\sigma_\phi^2 = 1.6$ (orange) they fan out and cross.

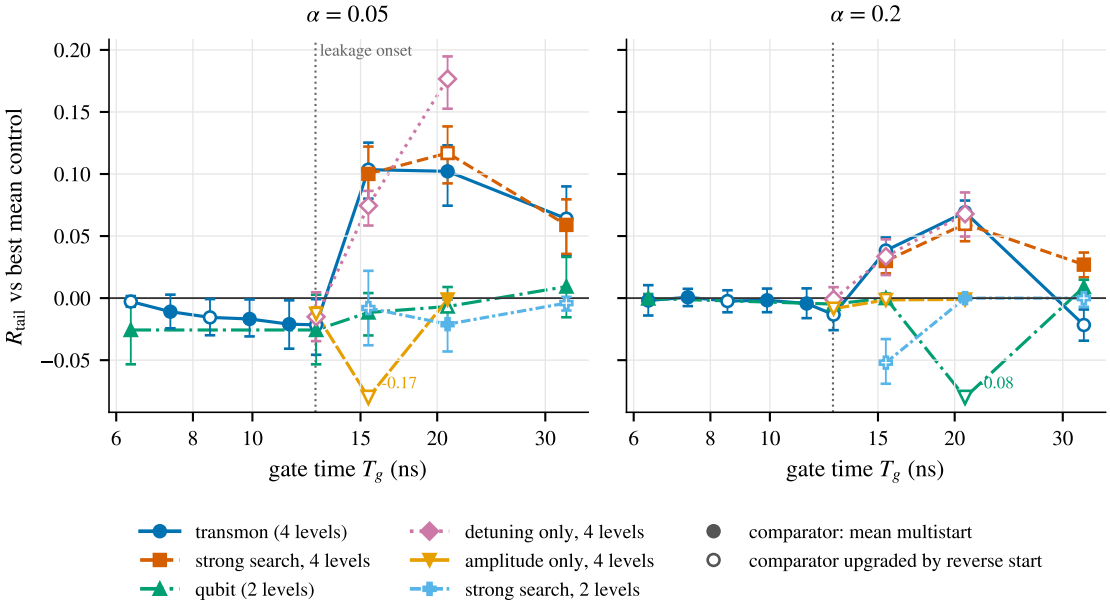



FIG. 5. $R_{\mathrm{tail}}$ versus gate time with detuning ($\sigma_\delta/2\pi = 2.5$ MHz) and amplitude ($\sigma_\gamma = 2\%$) noise. Dotted: leakage onset. Open markers: comparator upgraded by a reverse start; values below the axis are marked at its lower edge. Throughout the leakage-dominated region the objectives coincide. Above it, the transmon diverges and the two-level qubit with the same noise does not, with 4 or 20 restarts; amplitude noise alone gives no positive divergence.

sent from the problem; both optimizers suppress it to $7$–$9 \times 10^{-5}$, and the residual loss distributions remain ordered. Where $R_{\mathrm{tail}}$ differs from zero at all it is slightly negative, the out-of-sample penalty of estimating the tail from six training draws.

Above the onset the transmon diverges and the qubit does not. With 20 restarts and 1000 iterations per objective, the transmon gives $R_{\mathrm{tail}} = 0.100$, 0.117 and 0.059 at 15.4, 20.8 and 32.5 ns, while the two-level qubit with identical noise gives $-0.008$, $-0.021$ and $-0.004$ ($\sigma_\phi^2 = 0.06$, 0.11 and 0.26). Splitting the channels shows that amplitude noise alone never produces a positive divergence; detuning noise alone does (up to 0.18 at 20.8 ns; Sec. S7.C). This excludes a mechanism in which the level-2 AC Stark shift converts amplitude error into a quadratic phase. The higher levels therefore lower and amplify the detuning-driven divergence, but on their own create none. The two-level qubit with both channels

shows no divergence at $\sigma_\phi^2 = 0.26$ and $T_g = 32.5$ ns, where the detuning-only scan would suggest a few percent; the collapse of Sec. IV A was tested only between 8 and 20 ns.

### D. Several channels at once

Proposition 2 allows two independent channels to separate the objectives even at linear order. On the qubit at $T_g = 12.7$ ns with $\sigma_\delta/2\pi = 2.5$ MHz we scanned the amplitude width up to $\sigma_\gamma = 12.7\%$ and the correlation $\rho$ between the channels (Sec. S7.E). With independent channels the objectives stay within a few percent of each other; one point ($\sigma_\gamma = 0.08$) is significant and replicates only weakly (+0.044, +0.028, +0.003 over three seeds), and the predicted geometric signature, a more isotropic loss for the CVaR control, appears in one seed of three. The two-axis mechanism is real but weak, because optimized controls can change the split of their error between axes only slightly.

Correlating the channels, which reduces the noise to a single direction and might be expected to restore equivalence, produces instead the largest divergences in this scan: $R_{\text{tail}} = 0.23$ at $\rho = -0.99$ and 0.29 at $\rho = +0.99$, with every reverse start returning to $u_{\text{mean}}$. The loss distributions cross [Fig. 6(c)]. The reason is that correlated noise lets the optimizer cancel one combined first-order error direction instead of two, and it does: the linear response along the dominant noise direction falls from 0.076 at $\rho = 0$ to 0.013–0.017 at $|\rho| = 0.99$, and the mean loss of $u_{\text{mean}}$ falls fourfold. What remains is second order, with $\kappa$ above one. Rank-one *noise* is not rank-one *linear error*: the optimization itself removes the linearity on which equivalence rests. Correlation makes the problem easier on average and harder in the tail.

### E. Mechanism: tail-targeted pre-compensation

*$\kappa$ orders the results.* A single quantity, the second-order ratio $\kappa$ of the mean control, orders the divergences of all experiments [Fig. 7(a)]. No point with $\kappa \lesssim 0.1$ shows a positive divergence. For a single channel $\kappa$ grows in proportion to $\sigma_\phi$ (0.075, 0.15, 0.25 and 0.5 at $\sigma_\phi^2 = 0.04$, 0.16, 0.41 and 1.6), so $\kappa \approx 0.1$ reproduces the threshold $\sigma_\phi^2 \approx 0.05$–$0.1$ of Sec. IV A. It also covers correlated channels, where the cancellation of the linear error raises $\kappa$ at fixed noise strength. The same threshold governs the whole control family: in the candidate pools, the fraction of rank reversals is ordered by $\kappa$ and switches on between $\kappa = 0.076$ and 0.13 [Fig. 7(b)]. The transmon diverges at somewhat smaller $\kappa$ (0.1–0.24), consistent with the leakage contribution that $\kappa$ omits.

*The gain is quadratic in $\kappa$, with the predicted coefficient.* Proposition 6 makes the threshold quantitative, and the saved error clouds test it directly. For each diagnosed mean control we evaluated the closed form (13) from its fitted $A$ and $Q$ and compared it with the numerically optimized reduced model [Fig. 8(b)]. Over the 30 controls with $\kappa < 0.25$ the ratio of prediction to model has median 1.07 and interquartile range 0.94–1.12, although their coefficients $C$ differ by a factor of twenty (0.22 to 4.53) and both one- and two-dimensional noise geometries are present; the residual 7% is the $O(\kappa)$ correction. The two-level detuning family has $C = 4.53$, nearly constant along the whole phase-variance scan, so its gain crosses the resolution of this study at $\kappa = \sqrt{0.04/4.53} = 0.094$ [Fig. 8(c)]. Measured: no point with $\kappa \leq 0.10$ shows a significant divergence, and none of the 42 candidate pools with $\kappa \leq 0.076$ shows a rank reversal, while reversals appear at every point with $\kappa \geq 0.13$. The empirical threshold of this paper is therefore a consequence of Theorem 1's hypothesis failing quadratically, not an independent finding.

One prediction is not confirmed. Equation (14) says the optimal offset should fall from 4.15 at small $\kappa$ to 2.76 at large $\kappa$, so the divergent points of the phase-variance scan ($\kappa \approx 0.5$) should sit near 3.4. They sit at 2.46–2.54, close to the $\kappa \to \infty$ value instead. The fixed-geometry model is the reason: a real optimizer also reshapes $A$ and $Q$ rather than moving the offset alone, and it estimates the tail from six training draws. The offset prediction is therefore quantitative only in the limit, while the gain prediction (13) is accurate where it is asymptotically valid.

*The reduced model predicts the size.* $\kappa$ is necessary but not sufficient. At $\kappa \approx 0.5$ detuning noise at $\sigma_\phi^2 = 1.6$ gives $R_{\text{tail}} = 0.275$, while independent channels at $\sigma_\gamma = 0.127$ give 0.017. The difference is geometric: with one dominant noise direction the second-order error of every tail draw points the same way and one offset cancels it; with two independent directions it does not. Proposition 5 retains this geometry. Computed from each mean control's fitted $A$ and $Q$ with no free parameter, $R_{0.05}^{\text{model}}$ resolves the case above (0.42 against 0.06–0.08) and orders the measurements with Spearman correlation 0.89 [Fig. 7(c)], against 0.83 for $\kappa$ alone on the same points. It acts as an upper envelope: measured values reach 6–87% of it, and the two points above it exceed it by less than their confidence intervals. The shortfall is expected, since the Lindblad floor dilutes $R_{\text{tail}}$ and the real optimizer compares two optimized controls rather than two offsets.

Once $\kappa$ is known, the experimental parameters add no further information (Sec. S7.F). The correlation $|\rho|$ predicts $R_{\text{tail}}$ on its own (Spearman 0.70) but has partial correlation $-0.00$ ($p = 0.98$) given $\kappa$; the same holds for the noise strengths. Gate time retains a borderline residual given $\kappa$ ($p = 0.05$) that disappears given $R^{\text{model}}$ ($p = 0.45$). Within these data, correlation, gate time and noise strength act only through $\kappa$ and the error geometry, as Proposition 5 requires. The test is observational; the parameters were not varied independently, and the controlled experiment of Sec. IV F shows that this conclusion is too strong for $\kappa$ taken alone.

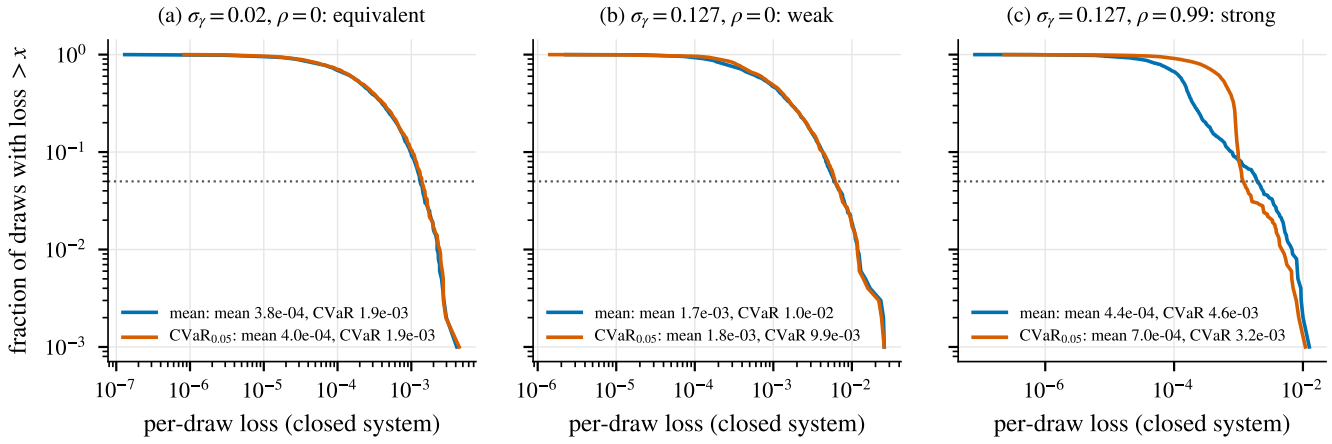



FIG. 6. Complementary cumulative distributions of the per-draw loss for the mean control (blue) and the $\mathrm{CVaR}_{0.05}$ control (orange), closed-system model, 1000 test draws. Dotted: the worst 5%. (a) Weak amplitude noise and (b) strong independent channels: the distributions coincide, in the tail as well as in the bulk. (c) Strongly correlated channels ($\rho = 0.99$): they *cross*. The CVaR control is worse in the bulk (mean loss $7.0 \times 10^{-4}$ against $4.4 \times 10^{-4}$) and better in the tail ($\mathrm{CVaR}_{0.05}$ of $3.2 \times 10^{-3}$ against $4.6 \times 10^{-3}$).

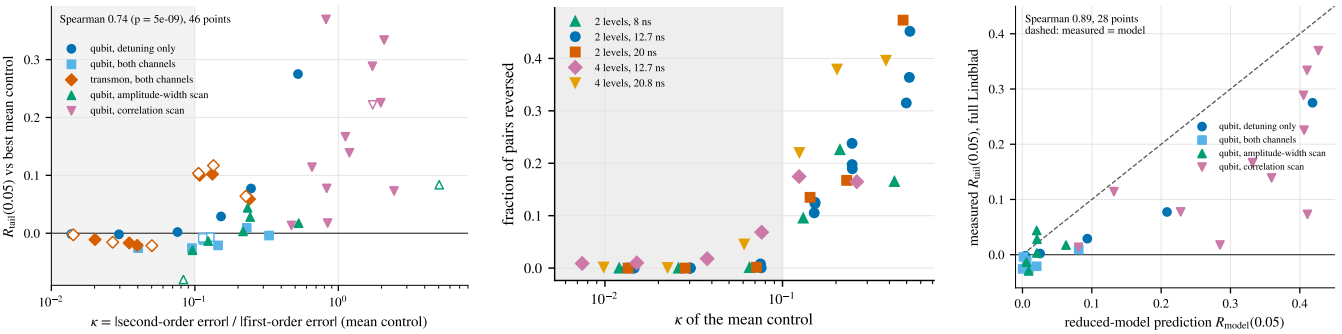



FIG. 7. One variable orders the divergence. (a) $R_{\mathrm{tail}}(0.05)$ against the second-order ratio $\kappa$ of the mean control, for every point with saved error clouds (Spearman 0.74, $p = 5 \times 10^{-9}$, 46 points). (b) Fraction of confirmed rank reversals in the candidate pools against $\kappa$ (Spearman 0.87, $p = 4 \times 10^{-14}$, 42 settings); on the qubit reversals are absent below $\kappa = 0.076$ and present above $\kappa = 0.13$. Shaded in (a,b): $\kappa < 0.1$. (c) Measured $R_{\mathrm{tail}}(0.05)$ against the parameter-free prediction $R^{\mathrm{model}}_{0.05}$ of Proposition 5 for two-level points (Spearman 0.89, $p = 1 \times 10^{-10}$, 28 points); dashed: equality. Open markers: comparator upgraded by a reverse start.

*A prospective test.* To test the model on data it had not seen, we fixed three predictions before running four new points (Table III): (a) $R_{\mathrm{tail}} \le R^{\mathrm{model}}$ at every point; (b) $R_{\mathrm{tail}}$ ranks as $R^{\mathrm{model}}$; (c) in the noise scan, $\kappa$ is proportional to the noise. Predictions (a) and (b) hold at all four points, and the ranking is exact. Prediction (c) fails ($\kappa \propto s^{0.5}$ over $s = 0.25$–$1$). It misapplied Proposition 5(ii), which holds for a *fixed* control: the mean control is re-optimized at each noise level and cancels more of its linear error at lower noise, so $\kappa$ falls more slowly than the noise. The 8 ns point gives the largest divergence of the study, $R_{\mathrm{tail}} = 0.37$.

TABLE III. Prospective test at $\rho = 0.99$, two levels: four points run after the predictions were fixed. "Fast": closed-system value, without the $T_1/T_2$ floor. °Comparator upgraded by a reverse start.

| point | $\kappa$ | $R^{\mathrm{model}}_{0.05}$ | $R_{\mathrm{tail}}(0.05)$ | fast |
|---|---|---|---|---|
| 12.7 ns, noise ×0.25 | 0.84 | 0.285 | 0.018(8) | 0.156 |
| 12.7 ns, noise ×0.5 | 1.20 | 0.359 | 0.14(4) | 0.247 |
| 20 ns, matched $\sigma^2_\phi$ | 1.74 | 0.411 | 0.22(3)° | 0.264 |
| 8 ns, matched $\sigma^2_\phi$ | 0.82 | 0.427 | 0.37(3) | 0.377 |
| reference, 12.7 ns | 1.73 | 0.405 | 0.29(5) | 0.305 |

*What CVaR does.* The optimized controls behave as Remark 1 predicts [Fig. 9(c)]. At every divergent point every CVaR control carries a coherent error antiparallel to its second-order error (cosine −1.00). Across the phase-variance scan its size is 2.46–2.54 $|\mathbf{q}|$ at $\alpha = 0.05$ and 1.61–1.71 $|\mathbf{q}|$ at $\alpha = 0.2$, against 2.76 and 1.70 pre-

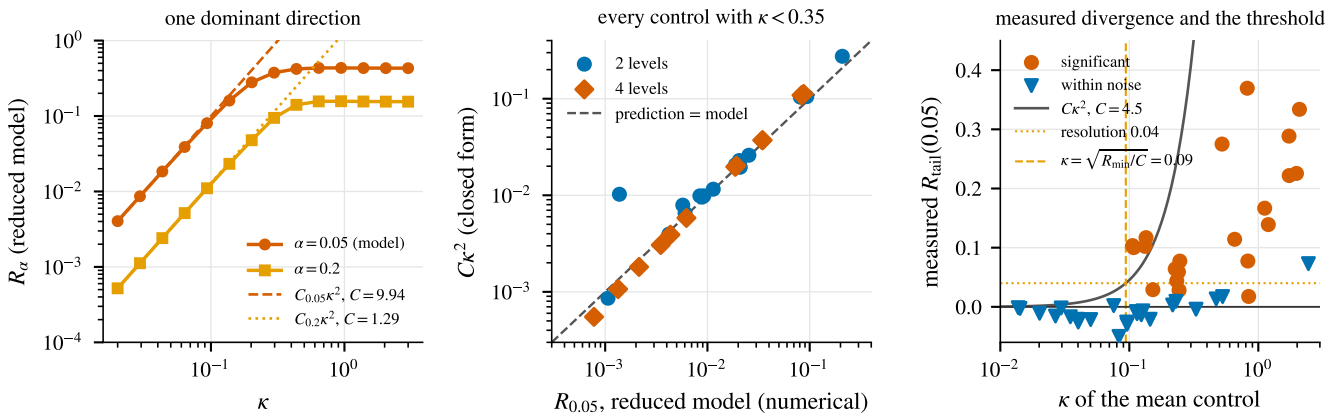



FIG. 8. The $\kappa^2$ law. (a) Reduced model with one dominant noise direction: the relative tail gain against $\kappa$ (markers) and the closed-form law $C_\alpha\kappa^2$ of Proposition 6 (lines), with no fitted parameter. The gain saturates at the pre-compensation limit of Remark 1. (b) The closed form against the numerically optimized reduced model for every diagnosed control with $\kappa < 0.35$; the two agree over three decades (median ratio 1.07, interquartile range 0.94–1.12 over 30 controls whose coefficients $C$ span 0.22–4.53). (c) Measured $R_{\text{tail}}(0.05)$ against $\kappa$, with the law for the two-level detuning family ($C = 4.5$) and the resolution of the study (0.04). Their intersection, $\kappa = \sqrt{R_{\min}/C} = 0.094$, falls between the largest $\kappa$ at which no rank reversal is observed (0.076) and the smallest at which reversals appear (0.13).

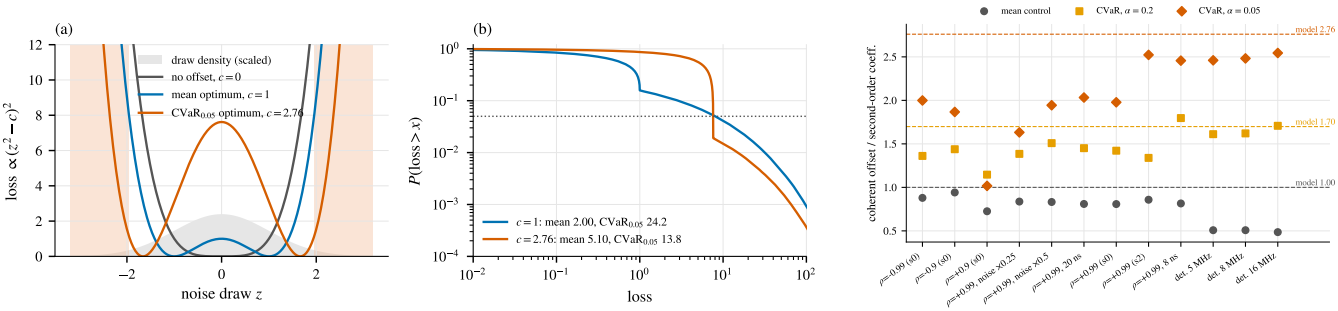



FIG. 9. Pre-compensation. (a) Per-draw loss $(z^2 - c)^2$ of the one-dimensional model of Remark 1; shaded: the worst 5% of draws. The mean optimum ($c = 1$) cancels the error at $|z| = 1$; the CVaR optimum ($c = 2.76$) cancels it at $|z| \approx 1.66$ and accepts a large error at $z = 0$. (b) The resulting distributions cross near the 5% level: the CVaR choice has 2.5 times the mean loss but a 43% lower $\text{CVaR}_{0.05}$. (c) Measured coherent offset $c = -\mathbf{e}_0 \cdot \mathbf{q}/|\mathbf{q}|^2$ of the optimized controls at every significantly divergent point where one noise direction dominates. Dashed: the parameter-free predictions 1, 1.70 and 2.76.

dicted. The mean controls sit near $c = 1$ when their linear error is cancelled (0.73–0.94 in the correlated-channel points) and near 0.5 in the single-channel scan, where a large linear error remains. The shortfall at $\alpha = 0.05$ is expected: the optimizer can also reshape $A$ and $Q$, and it estimates the tail from six draws. The same signature marks every divergent point: the nominal (noise-free) loss is a much larger share of the mean loss for the CVaR control than for $u_{\text{mean}}$ (0.83 against 0.08 at $\sigma_\phi^2 = 1.6$), while in the equivalence region the two agree. CVaR does not find a lower-error pulse. It trades nominal fidelity for a flatter tail, by pre-compensating the second-order error of the worst draws.

### F. A controlled test of the mechanism variable

The evidence so far is observational: the experimental parameters were never varied independently of $\kappa$. We therefore ran a designed experiment. For each cell of a $3 \times 3$ grid in $(\rho, T_g)$ we *solved* for the noise scale that puts the mean control's $\kappa$ on a target value $\kappa^\star \in \{0.20, 0.35, 0.55\}$, by secant iteration in $\log s$ over the measured $\kappa(s)$ (Sec. S9.A), and then ran the standard comparison there with two noise seeds: 27 cells, 54 points, all two-level. Rows of the design differ in $\kappa$ and columns differ in the knobs at approximately fixed $\kappa$.

*$\kappa$ orders the outcome.* Median $R_{\text{tail}}$ rises across the rows, +0.004, +0.037, +0.080 (cells above the $T_1/T_2$ floor), and Spearman$(\kappa, R_{\text{tail}}) = +0.49$ over all 54 points.

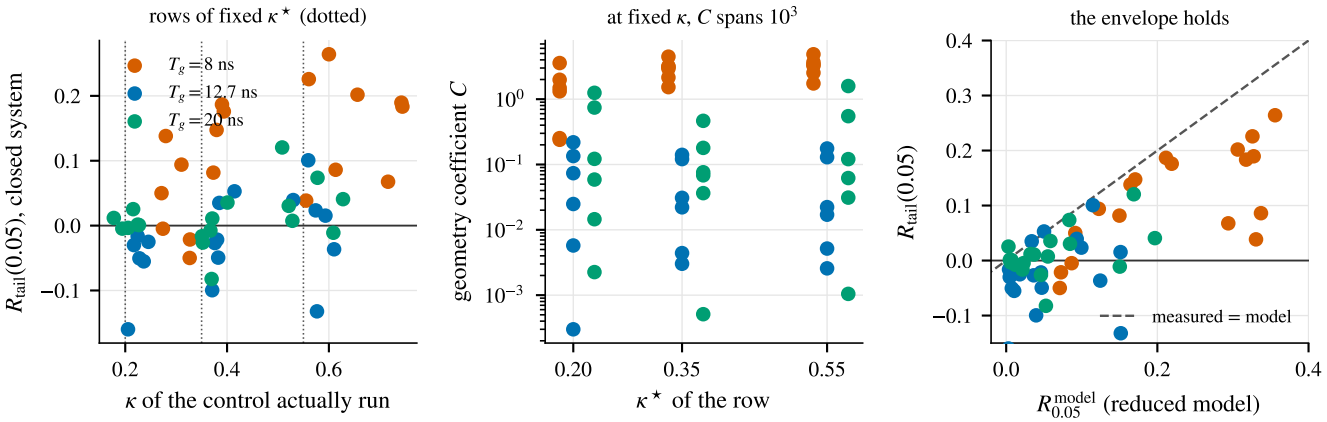



FIG. 10. The controlled experiment at fixed $\kappa$. (a) $R_{\text{tail}}(0.05)$ against the $\kappa$ of the control actually run; dotted lines mark the three target values. The rows separate, but within each row the gate times do not overlap. (b) The closed-form coefficient $C$ of Proposition 6 at each target: at fixed $\kappa$ it spans three orders of magnitude, and the split is by gate time. (c) Measured against the reduced-model prediction; the envelope holds at 51 of 54 points.

*$\kappa$ does not explain it.* At fixed $\kappa$, gate time still predicts the gain: the partial rank correlation of $R_{\text{tail}}$ with $T_g$ given $\kappa$ is $-0.33$ ($p = 0.015$), and $-0.58$ ($p < 0.001$) among the cells whose mean loss is above the dissipative floor. The claim of Sec. IV E, that the knobs act on the divergence only through $\kappa$, does not survive a controlled design; it was an artefact of how the observational points happened to be distributed.

*The geometry is what is missing.* At fixed $\kappa$ the closed-form coefficient $C$ of Proposition 6 varies by three orders of magnitude, and it varies *with gate time*: median $C = 1.4$, 3.0, 3.3 at $T_g = 8$ ns for the three rows, against 0.02–0.09 at 12.7 and 20 ns [Fig. 10(b)]. A $\kappa$ of 0.35 at 8 ns and a $\kappa$ of 0.35 at 20 ns are not the same physical situation: in the first the second-order error is aligned with the dominant noise direction, in the second it is not. Replacing $\kappa$ by the reduced model, which keeps that geometry, removes the gate-time dependence entirely: the partial correlation of $R_{\text{tail}}$ with $T_g$ given $R^{\text{model}}$ is $+0.01$ ($p = 0.95$), and $-0.04$ ($p = 0.84$) above the floor, while Spearman$(R^{\text{model}}, R_{\text{tail}}) = +0.81$ there. The correlation $\rho$ shows a residual effect given $R^{\text{model}}$ over the full set ($-0.29$, $p = 0.03$) which disappears above the floor ($-0.12$, $p = 0.49$): it comes from the $\rho = 0.9$ cells, whose mean losses of $\sim 10^{-5}$ leave the CVaR estimate dominated by its six training draws, and six of those eighteen cells come out significantly *negative*.

*Caveats of the design.* The calibration fixes $\kappa$ of the control it optimizes, but the run's mean control is better optimized and cancels more linear error, so its $\kappa$ is larger by a median factor 1.08 — 1.30 at 8 ns. The rows therefore overlap (0.18–0.33, 0.35–0.56, 0.51–0.74) and every analysis above uses the measured $\kappa$, not the target. The envelope property survives the design: three of 54 points sit above $R^{\text{model}}$, none beyond its confidence interval.

### G. Leakage-aware $\kappa$

$\kappa$ is read from the qubit block of the error and is therefore blind to leakage, which is why the transmon diverges at a smaller $\kappa$ than the qubit. We extend the error coordinate to the leakage amplitudes, $x = (\boldsymbol{\epsilon}; \operatorname{Re} l; \operatorname{Im} l)$ with $l_{kj} = \langle k|U|j\rangle$ for $k \geq 2$ in the same global-phase gauge, and carry the loss metric with it: to second order $L \approx x^{\mathsf{T}} W x$ with $W = \operatorname{diag}(W_\epsilon, W_{\text{leak}})$ and $W_{\text{leak}} = \frac{1+\lambda}{4}\left[\begin{smallmatrix} I\otimes G_r & -I\otimes G_i \\ I\otimes G_i & I\otimes G_r \end{smallmatrix}\right]$, $G = \overline{\sum_\psi |\psi\rangle\langle\psi|}$ (Sec. S9.B). The mechanism variable becomes $\kappa_{\text{lk}} = \|W^{1/2}Q\|_F / \|W^{1/2}A\|_F$ in these coordinates.

Re-diagnosing every saved control of every experiment with these coordinates (no new optimization) gives three results. The decision thresholds of the two models move together: fitted separately, the threshold is 0.387 for the qubit and 0.055 for the transmon in $\kappa$, a factor of 7.0, against 0.248 and 0.142 in $\kappa_{\text{lk}}$, a factor of 1.7 [Fig. 11(a)]. The transmon is classified better (AUC 0.949 → 0.978; pooled 0.732 → 0.757). And the leakage-aware reduced model is an envelope for the transmon at 32 of 33 points, against 22 of 33 for the qubit-only model. The reason is visible in the error budget: leakage is a median 0.3% of the first-order response but 9.5% (up to 23%) of the second-order term [Fig. 11(c)] — exactly the quantity $\kappa$ is meant to measure. The one remaining violation is the four-level point at $T_g = 20.8$ ns and $\sigma_\phi^2 = 0.41$, where $R_{\text{tail}} = +0.274$ against a bound of 0.210 with a confidence interval that excludes it.

### H. Generality: angle, anharmonicity and pulse basis

Finally we varied the things the study had held fixed: the target rotation angle ($\pi/4$, $\pi/2$, $\pi$), the anharmonic-

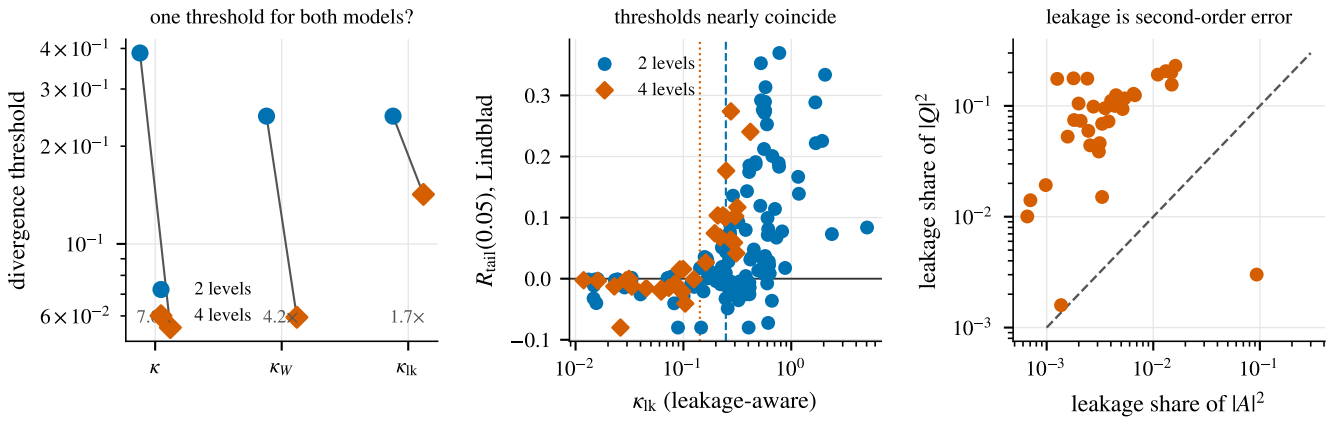



FIG. 11. Leakage-aware $\kappa$, over all 165 diagnosed mean controls (132 two-level, 33 four-level, 75 of them significantly divergent). (a) Divergence threshold fitted separately for the two models; the factor between them falls from 7.0 to 1.7. (b) $R_{\mathrm{tail}}$ against $\kappa_{\mathrm{lk}}$, with the two thresholds. (c) Leakage contributes a median 0.3% of $\|A\|^2$ but 9.5% of $\|Q\|^2$: it is almost entirely second-order error.

ity ($-150$, $-200$, $-300$ MHz, four levels) and the pulse basis (17 sine modes, 6 sine modes, and 17 piecewise-constant segments, the last matching the parameter count of the first), each at two to four noise scales; 22 points (Sec. S9.C). Note that with an $x/y$ drive and no $z$ control a Hadamard is $X_{\pi/2}$ up to virtual-$Z$ frames, so the physically distinct axis is the rotation angle, which sets the Rabi rate and hence $\delta/\Omega$.

Over these points Spearman$(\kappa, R_{\mathrm{tail}}) = +0.82$ ($p = 2\times10^{-5}$) and Spearman$(R^{\mathrm{model}}, R_{\mathrm{tail}}) = +0.86$, with no detectable residual dependence on anharmonicity ($p = 0.24$) or basis ($p = 0.84$). The anharmonicity scan is the clean null: $\kappa = 0.035$–$0.099$ across $-150$ to $-300$ MHz with $R_{\mathrm{tail}}$ between $-0.012$ and $+0.030$, every point inside its envelope. The $\pi$ pulse behaves as the mechanism predicts and as spin echo would suggest: it refocuses static detuning to first order, its linear response nearly vanishes, $\kappa$ reaches 0.8–9.6 and $R_{\mathrm{tail}}$ reaches 0.28–0.55 — but at mean losses of $10^{-8}$–$10^{-5}$, far below the $T_1/T_2$ floor, so the effect is real in the closed system and invisible on hardware.

Two failures are worth recording. The piecewise-constant basis, at matched parameter count, makes the optimization unreliable: at the nominal noise its mean multistart misses a basin that the reverse start finds ($R_{\mathrm{tail}} = -0.305$ against the recovered comparator), and at twice the noise it exceeds its own envelope ($+0.132$ against 0.048). This is a conditioning failure of the search, not a property of the physics, and it means the mechanism test is not portable to that basis without a better-conditioned parameterization. Second, the raw comparison at these points was initially misleading for a mundane reason: the closed-system $R_{\mathrm{tail}}$ stored by the pipeline is measured against the multistart mean control, and the best-mean comparator of Sec. III is applied by the Lindblad validation step, which cannot run for a non-standard target. Applying the comparator rule to the saved closed-system losses changed 46 of 76 Stage-3 points, by more than 0.05 at four of them, and removed a spurious $R_{\mathrm{tail}} = +0.31$ at a setting where Sec. IV A reports zero. It is the same pitfall the protocol was built for, arriving through a new door.

## I. Colored noise: memory, and the effective dimension

Every experiment so far has used quasi-static noise: one random draw held fixed for the duration of the gate. Real dephasing has memory. Corollary 1 says memory should change the size of $\sigma_\phi^2$ and nothing else, which is a prediction the study had never tested. We replaced the static detuning by an Ornstein–Uhlenbeck process of correlation time $\tau_c$, sampled exactly as a stationary AR(1) sequence on the control grid and normalized so that the accumulated phase variance $\mathrm{Var}(\phi) = \sigma_{\mathrm{ou}}^2\, dt^2 \sum_{jk} e^{-|t_j - t_k|/\tau_c}$ matches the quasi-static $\sigma_\phi^2$ at every $\tau_c$. That normalization is what makes the comparison fair: the arms differ in the *shape* of the noise, not its integrated strength. The grid is $\tau_c/T_g \in \{0.1, 0.3, 1, 3, 10\}$ plus a quasi-static arm, crossed with $\sigma_\phi^2 \in \{0.04, 0.16, 0.41\}$ and two seeds: 36 points, all at $T_g = 12.7$ ns on the two-level model, with the best-mean comparator applied at save time (it upgraded 20 of the 36, by at most 0.064). Full protocol in Sec. S10.

*The control arm reproduces the static study.* The colored propagator is a separate code path from the one used in Secs. IV A–IV H, so its quasi-static limit is a check on both. It gives $C = 4.523$–$4.538$ across the six quasi-static points against the 4.53 of the detuning family in Sec. IV E, and $R_{\mathrm{tail}} = +0.028$ at $\sigma_\phi^2 = 0.16$ against the $+0.029$ of Sec. IV A.

*Memory does not move the comparison.* Across the whole grid $|R_{\mathrm{tail}}| \le 0.043$, and the raw rank correlation

of $R_{\rm tail}$ with $\log \tau_c$ is $+0.25$ ($p = 0.14$), $+0.27$ controlling for $\sigma_\phi^2$ ($p = 0.12$) [Fig. 12(a)]. Corollary 1 does not claim that $\tau_c$ is invisible, however — it claims that $\tau_c$ acts *only* by rescaling the effective noise, that is, only through $\kappa$ and the geometry. The sharp test is therefore the one used in Sec. IV F: control for the reduced model as well. Doing so removes the dependence entirely, to $+0.07$ given $\kappa$ ($p = 0.70$), $+0.10$ given $C\kappa^2$ ($p = 0.59$) and $+0.05$ given $R^{\rm model}$ ($p = 0.77$) [Fig. 12(c)]. The same holds inside the only row where $R_{\rm tail}$ is large enough to resolve: at $\sigma_\phi^2 = 0.41$, $R_{\rm tail}$ does fall with decreasing $\tau_c$ (Spearman $= +0.65$, $p = 0.02$), but it tracks $C\kappa^2$ while doing so ($+0.67$, $p = 0.02$), and the residual dependence on $\tau_c$ given $C\kappa^2$ is $+0.21$ ($p = 0.54$). Colored noise moves the mechanism variable, and the mechanism variable moves the gain.

*What the pre-registered mechanism test can and cannot show.* The two seeds put the run-to-run standard deviation of $R_{\rm tail}$ at 0.021 against a total spread of 0.026, so the reproducible signal has standard deviation 0.015 and *any* correlation with $R_{\rm tail}$ is attenuated by at most $(1 + \sigma^2_{\rm noise}/\sigma^2_{\rm signal})^{-1/2} = 0.60$. The pre-registered threshold for this stage was 0.6, which therefore demanded a perfect underlying correlation and could not be met. Against the ceiling, the observed values are at it: Pearson$(R^{\rm model}, R_{\rm tail}) = 0.67$, a regression slope of $0.30 \pm 0.06$ — properly inside $(0, 1]$ for a bound — and a residual scatter of 0.019, which is 0.93 times the scatter between repeats of the same cell [Fig. 12(b)]. The reduced model leaves less unexplained variation than rerunning the experiment does. One point of 36 exceeds its $C\kappa^2$ envelope.

*Color raises the effective dimension.* The tail-to-mean ratio is 5.36 in the quasi-static arm, close to the $r_\alpha = 5.58$ of Proposition 1, and falls monotonically to 3.44 at $\tau_c = 0.1\,T_g$, passing through the 4.00 of Proposition 2 on the way. Read through Proposition 3 this is not a failure of the shape prediction but a measurement of $m_{\rm eff}$, which rises from 1.09 to 2.8 [Fig. 13(b)]. The mechanism is visible directly: a control sees colored noise through more than the accumulated phase, and the share of $\|A\|^2$ carried by the first-moment coordinate $z_2$ climbs from 0 under quasi-static noise to 0.70–0.84 at short $\tau_c$ (Spearman$(\log \tau_c,\, z_2$ share$) = -0.72$, $p = 8 \times 10^{-6}$).

The prediction can be made per control rather than per row. Taking the two noise coordinates' shares of $\|A\|^2$ as the weights of Eq. (5) gives $\rho_\alpha(m_{\rm eff})$ from the control's geometry alone, before any tail is examined. Where two coordinates actually span the loss ($R^2_{\rm fit} \geq 0.94$, 18 points) the prediction matches the measured ratio to a median of 2.6%, with Spearman $= +0.89$ ($p = 8\times10^{-7}$) [Fig. 13(a)]. Where they do not ($R^2_{\rm fit}$ as low as 0.010 at $\tau_c \leq 0.3\,T_g$), the measured $m_{\rm eff}$ exceeds the two-coordinate participation ratio in 18 of 18 cases ($p = 4\times10^{-6}$): the truncation fails, and it fails in the one direction a truncation is allowed to fail. We regard that signed failure as better evidence for Eq. (6) than uniform agreement would have been.

Two limitations follow. At $\tau_c \leq 0.3\,T_g$ the two-coordinate reduction breaks down, so $C$ and $R^{\rm model}$ are not trustworthy there — which is also where $R_{\rm tail} \approx 0$, so little is at stake. And because the tail amplification falls with $m_{\rm eff}$, fast noise is doubly unfavourable to a tail objective: it lowers both the ratio that the objective exploits and, through the fitted geometry, the coefficient $C$, which drops from 4.53 quasi-static to 0.04–1.8 at $\tau_c \leq 3\,T_g$. A device whose dephasing is fast on the gate scale is a device where CVaR has least to offer.

### J. The same curve along a second route

Section IV I raised $m_{\rm eff}$ by giving the noise memory. Nothing in Proposition 3 refers to memory, so the sharper test is to raise $m_{\rm eff}$ by a mechanism with no time structure at all and ask whether the ratio lands on the same curve. We did that by rotating the noise between two physically distinct channels: a detuning of variance $\sigma_\delta^2 \cos^2\theta$ and a drive-amplitude error of variance $\sigma_a^2 \sin^2\theta$, with $\theta$ stepped in $11.25^\circ$ increments from pure dephasing to pure amplitude noise and the overall scale recalibrated at every $\theta$ to hold the mean loss fixed. Two seeds, $T_g = 12.7$ ns, two-level: 18 points. Full protocol in Sec. S11.

*The law holds, and the pre-registered statistic did not see it.* Within a seed the agreement is close to exact: the measured ratio falls monotonically as the participation ratio rises, with Spearman$(m_{\rm eff}, \text{ratio}) = -0.976$ and $-0.952$ against a predicted $-1$, and Spearman of prediction against measurement of $+0.976$ ($p = 3 \times 10^{-5}$) and $+0.952$ ($p = 3 \times 10^{-4}$), at median errors of 7.9% and 3.4% [Fig. 14(a)]. Pooled across seeds the same statistic is $+0.712$, and the pre-registered threshold was 0.8. The difference is a nuisance level shift: seed 1 sits $+0.563 \pm 0.141$ above seed 0 in measured ratio at every $\theta$, while the prediction varies by only 1.51 over the whole range of $m_{\rm eff}$, so an offset of that size reorders the pooled ranking without touching the law. Blocking on seed — an elementary precaution we should have pre-registered and did not — gives a seed-centred Spearman of $+0.962$ ($p = 3 \times 10^{-9}$). We report the pooled failure and the blocked pass together; the second is the scientifically meaningful one and the first is a lesson about how to pre-register.

*One curve, two mechanisms.* The two routes can now be compared directly. Taking each point's residual from $\rho_\alpha(m_{\rm eff})$, the colored-noise route gives a median of $-0.1\%$ with 10th–90th percentiles $[-7.4, +2.6]\%$ over 18 points, and the channel-geometry route a median of $-1.5\%$ with $[-8.0, +3.9]\%$ over 16 [Fig. 14(b)]. A Mann–Whitney test does not distinguish them ($p = 0.77$). Pooling all 34 points against the single curve, with no cross-calibration and no fitted constant, gives Spearman $= +0.831$ ($p = 1 \times 10^{-9}$) at a median absolute residual of 3.7% [Fig. 14(c)], over closely matched coverage ($m_{\rm eff} \in [1.00, 1.99]$ and $[1.00, 1.92]$ respectively). Two mechanisms with nothing physical in common — one

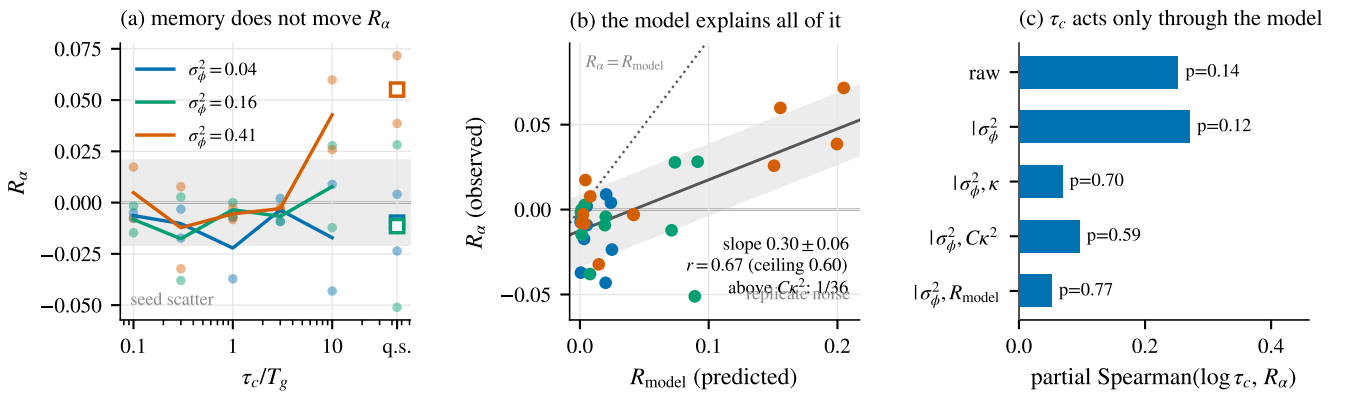



FIG. 12. Colored noise. (a) $R_{\text{tail}}$ against the noise correlation time at three phase variances, with the quasi-static limit at the right; lines are seed means, faint markers the individual runs, and the band is the run-to-run scatter. (b) $R_{\text{tail}}$ against the reduced-model prediction; the dotted line is equality, the solid line the fit, and the band the replicate noise, which the residual scatter already matches. (c) Partial rank correlation of $R_{\text{tail}}$ with $\log \tau_c$ under successive controls: the dependence on correlation time disappears once the mechanism variable is held fixed, as Corollary 1 requires.

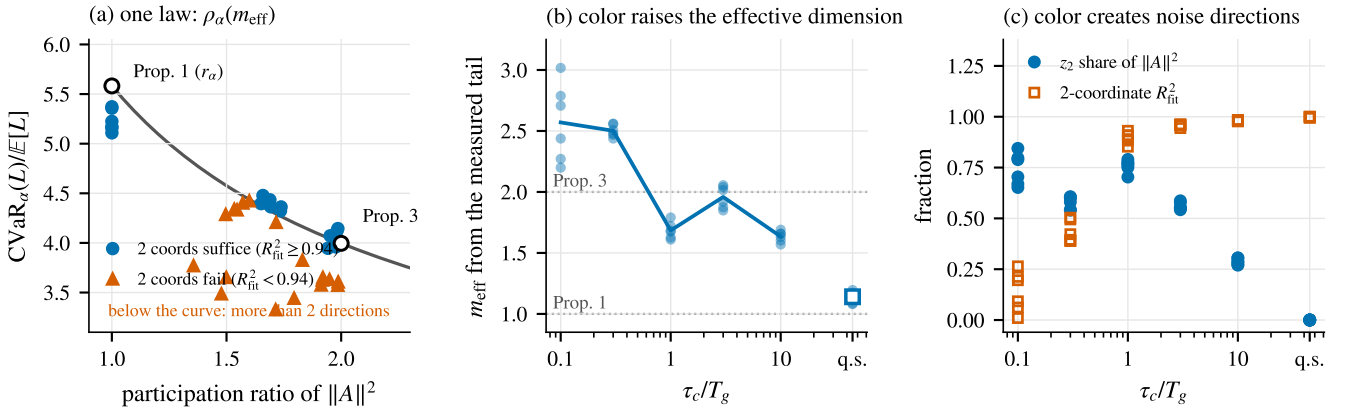



FIG. 13. The effective dimension. (a) Measured tail-to-mean ratio against the participation ratio of the control's linear response; the curve is $\rho_\alpha(m_{\text{eff}})$ of Proposition 3 and the two open circles are Propositions 1 and 2. Points for which two noise coordinates span the loss lie on the curve; points for which they do not lie below it, because the truncation undercounts directions. (b) $m_{\text{eff}}$ inverted from the measured tail, against correlation time. (c) The share of $\|A\|^2$ carried by the first-moment noise coordinate, and the quality of the two-coordinate fit.

temporal, one geometric — put the tail-to-mean ratio on the same one-parameter curve. That is the evidence that Proposition 3 describes a property of the loss rather than of the experiment that suggested it.

*Two tests that did not resolve.* The design produced only one pair separated by $\Delta m_{\text{eff}} > 0.3$, so the pre-registered dominance-crossing test had no power and we do not score it. The pre-registered $\kappa^2$-envelope test fails on 18 points and passes on the 16 in regime: the two excluded points are at $\theta = 90°$, pure amplitude noise, where the control nulls the linear response almost completely — mean losses of $5\text{–}6\times10^{-6}$ and $\kappa = 6.3$ and 46.9. A bound of the form $R = C\kappa^2 + O(\kappa^3)$ says nothing at $\kappa = 47$. On the remaining 16, all with $\kappa \leq 0.12$, the largest exceedance is 0.027 against a threshold of 0.10.

### K. Rare catastrophic channels: weight without leverage

Amplitude damping breaks (A1) in the way the theory is least equipped for: a single trajectory either relaxes or does not, and the loss when it does is set by the gate's structure rather than by a noise amplitude. We evaluated it exactly rather than by trajectory sampling. Expanding the damped propagator in the number of jumps and truncating after one gives a no-jump branch under the Strang-split map $K_k = D^{1/2}U_k D^{1/2}$ and a one-jump branch whose observable is accumulated by a single backward recursion, so the whole loss distribution — every branch, with exact weights — costs one extra pass and carries error $O((\Gamma T)^2)$. Two correct-

ness subtleties are what make it agree with the Lindblad channel: the branch probabilities differ per input state, so the four-state average must be Bayes-weighted, and the jump weight must be the telescoping trace loss of the discretized map rather than $\Gamma\, dt\langle n\rangle$. With both, the residual mass defect over the whole study is at most $2.2 \times 10^{-6}$. Derivation and validation in Sec. S12.

The grid crosses $T_g \in \{12.7, 40\}$ ns, $T_1 \in \{10, 30\}\,\mu$s and $\sigma_\phi^2 \in \{0.04, 0.16\}$ with two seeds, plus a four-level arm: 14 points, with $\beta$ spanning 0.098 to 63 and $p \leq 2.0 \times 10^{-3}$.

*The mixture law holds to one percent.* Across the twelve two-level points the measured tail-to-mean ratio ranges from 8.20 to 19.87 and Eq. (8) predicts it to a median of 1.2%, with every point above $r_\alpha$ as required [Fig. 15(a)]. We stress what is and is not being tested. The rank correlation between prediction and measurement is +0.991, but the predicted ratio is a monotone function of $\beta$ alone, so that number reproduces Spearman($\beta$, ratio) to machine precision and is not an independent check; the content of the test is the agreement in *level*, at no free parameters, across a factor of 640 in $\beta$. The four-level arm is the one place the law degrades, to 9.9% and 16.7%, in both cases with the measurement below the prediction — the signed direction expected when leakage adds a third channel that a two-component mixture does not carry.

*And the optimizer ignores it.* A jump share of 63 means the rare branch supplies 98% of the mean loss, and Eq. (9) says a tail objective weights it 3.58 times more heavily still. Neither fact moves the selected pulse. Over the 14 points $|R_{\rm tail}| \leq 0.13$ with a largest positive value of 0.030, and Spearman($\beta, R_{\rm tail}$) = +0.22 ($p = 0.45$) [Fig. 15(c)]. That null survives every subset we tried: the twelve two-level points (+0.11), dropping the two points with $\beta > 10$ (+0.17), and scoring against the raw mean comparator instead of the best one (+0.01). As in Sec. IV I the replicate noise caps what any such correlation can be — run-to-run standard deviation 0.032 against a total spread of 0.040, a ceiling of 0.62 — so the honest reading is a disattenuated +0.36, still far from significance. The companion prediction, that $\kappa$ becomes inert once $\beta$ is present, does hold (Spearman($\kappa, R_{\rm tail}$) = +0.25, $p = 0.39$; partial given $\beta$, +0.18, $p = 0.56$), but a null on both variables is a weaker statement than the substitution we predicted.

*Weight is not leverage.* The reason is visible in how little of the jump channel the control can reach. Across the eight controls optimized at each point — free, mean, three tail levels and three reverse warm starts — the jump loss $\ell_j$ spans a median of 0.9% and the integrated excited population $n_{\rm int}$, which sets $p$, a median of 1.4%, while the dephasing loss they are traded against spans 45% [Fig. 15(b)]. The gate's own requirement pins the excited population to within a percent, and the cost of a relaxation event is a property of the target unitary, not of the pulse. A tail objective therefore multiplies the weight on a channel whose shape the control cannot change, and the extra weight has nowhere to act. This also explains why our pre-registered estimator for the amplification failed: it scored the ratio $(n_{\rm free} - n_{\rm CVaR})/(n_{\rm free} - n_{\rm mean})$ against $\Xi_\alpha = 3.58$ and returned 0.72, stable to 0.66 as the resolution gate is tightened from eleven points to five. That identity presumes $n_{\rm int}$ is the coordinate along which the trade-off is resolved, and the 1.4%-against-45% spread says it is not; the pulse moves almost entirely in directions that leave $n_{\rm int}$ alone. The sign agrees at 13 of 14 points and the magnitude means nothing, and we report the estimator as mis-specified rather than the prediction as falsified.

*A pulse that nulls the linear response.* Two of the $\sigma_\phi^2 = 0.16$ cells illustrate both halves at once. At one seed the optimizer found a basin with the linear phase response almost entirely cancelled: the dephasing loss drops by a factor of 73 against its sibling seed, the mean loss improves from $2.8 \times 10^{-3}$ to $1.4 \times 10^{-3}$, and because $\|A\|_F$ collapses, $\kappa$ rises from 0.13 to 6.2 and $\beta$ to 63. Every one of the eight controls at that point lands in the same basin, so it is a property of the problem and not of one restart. It is the most tail-dominated pulse in the study by a wide margin — and $R_{\rm tail}$ there is $-0.0008$. When the remaining loss is carried by a channel the pulse cannot shape, driving $\beta$ up does not create a tail-mean separation; it only moves the ratio along Eq. (8).

*What this adds to the scope.* Section IV E established that the tail gain is quadratically small in $\kappa$. This section adds a condition that $\kappa$ does not capture. A channel can dominate the mean loss, sit far out in the tail, and be amplified several-fold by the objective, and still not change the answer, because selection responds to the *gradient* of a channel with respect to the control and not to its weight in the objective. A rare catastrophic channel is therefore not by itself a reason to reach for a tail objective. The question to ask of a device is not how heavy its worst channel is but how much of it the pulse can move.

### L. Predicting the gain before paying for it

Everything so far diagnoses the gain after both optimizations have been run. For the result to be useful on hardware the order has to be reversed: one should be able to decide, from measurements on the mean-optimal pulse alone, whether a tail objective is worth the shots. This section builds that predictor and measures how well it works.

The construction is the reduced model of Proposition 5 read backwards. Let $\epsilon_k$ be the error coordinates of draw $k$ under the mean-optimal control and $L_k = \epsilon_k^{\mathsf{T}} W \epsilon_k$ its loss. The $\alpha$-tail selects a sub-population; its conditional mean displacement

$$d_\alpha \;=\; \mathbb{E}[\epsilon \,|\, L \geq \mathrm{VaR}_\alpha(L)] \;-\; \mathbb{E}[\epsilon] \tag{17}$$

is the direction in which the tail draws differ from the ensemble, and a common offset $-t\, d_\alpha$ applied to every

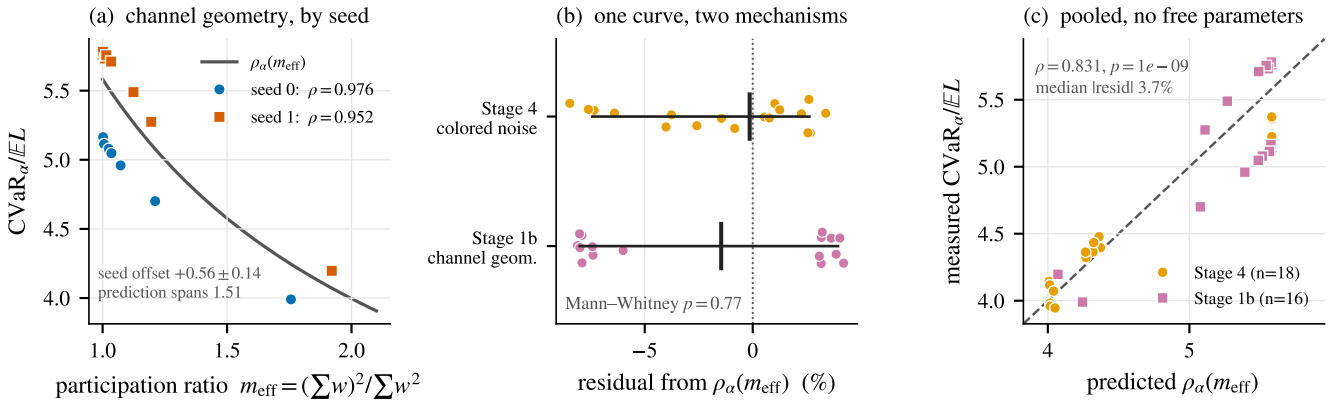



FIG. 14. The effective dimension along a second route. (a) Measured tail-to-mean ratio against the participation ratio as noise is rotated from pure dephasing to pure amplitude error, shown separately by seed; the curve is $\rho_\alpha(m_{\text{eff}})$ of Proposition 3. The two seeds track the curve with a constant offset between them, which is what defeats a pooled rank statistic. (b) Residuals from that curve for the colored-noise route of Sec. IV I and the channel-geometry route here; ticks are medians, bars the 10th–90th percentiles. (c) All 34 points against the prediction, with no free parameter and no cross-calibration.

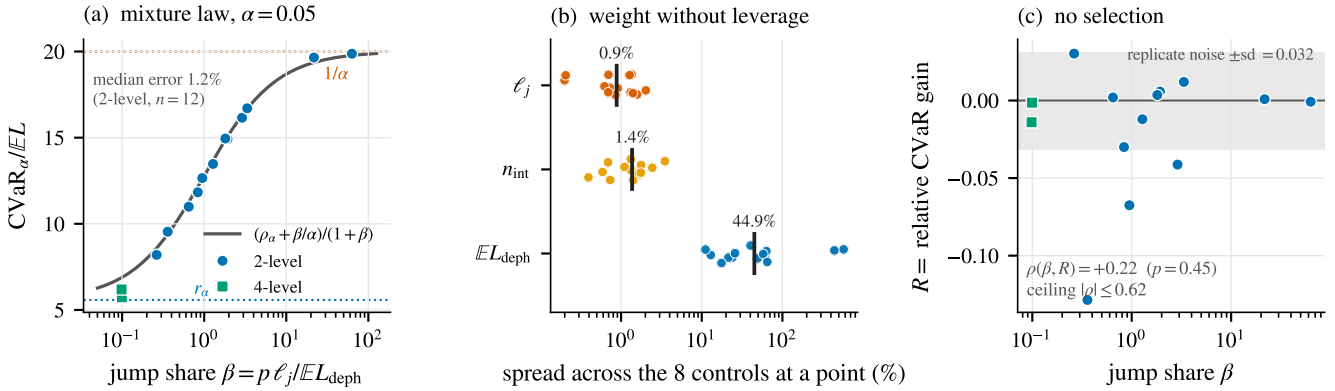



FIG. 15. A rare catastrophic channel. (a) Measured tail-to-mean ratio against the jump share $\beta$; the curve is Eq. (8) with no fitted parameter, and the dotted lines are the two limits $r_\alpha$ and $1/\alpha$. (b) How much of each quantity the control can reach: spread across the eight controls optimized at a point, for the jump loss, the integrated excited population that sets the jump probability, and the dephasing loss they are traded against. The jump channel is nearly immovable. (c) The tail gain against $\beta$, with the band showing the run-to-run scatter. Even at $\beta = 63$, where the rare branch carries 98% of the mean loss, the two objectives select the same pulse.

draw trades bulk loss for tail loss. The optimal $t$ follows from a one-dimensional line search over the measured losses, which requires no model of the control at all. Two refinements matter in practice and are derived in the Supplemental Material [30]: the curvature of $\text{CVaR}_\alpha$ in the offset is not $2W$ but carries a boundary term from draws entering and leaving the tail, and the step must be projected onto the reachable subspace $\mathcal{B} = \text{range}(\partial\epsilon/\partial c)$. We write the resulting prediction $R_{\text{pred}}$. Every ingredient—$\epsilon_k$, $L_k$, $W$, and the Jacobian—is measurable.

*A protocol correction.* Scoring $R_{\text{pred}}$ against a measured gain requires a mean-side comparator, and our first attempt at a fair one was wrong in a way worth recording. Following Sec. IV A we scored the CVaR control against the best of $\{u_{\text{mean}}\} \cup \{\text{reverse warm starts}\}$, a reverse warm start being a mean optimization initialized at a CVaR solution. In 10 of 16 cells at least one reverse run found a strictly *better mean optimum* than the mean multistart itself, by a factor of four in one cell. Where that happens the comparator is not a fairness correction: the mean arm has effectively had five restarts, four of them initialized with knowledge of where the good basins lie, against the CVaR arm's one. The symptom was unmistakable in hindsight—the raw gain $R^{\text{raw}}$ was positive at 61 of 64 point–$\alpha$ pairs while the comparator-based gain was negative at half of them.

The repair is symmetry. We pool every control any arm found at a point, deduplicate, and run *every* objective from *every* start in that pool, each arm reporting its best

as selected on the validation ensemble by its own risk measure, with the test ensemble touched only for the final evaluation. Equal budget, equal information, and no arm selects on the quantity being compared. The effect is large: the replicate scatter in $R$ between the two seeds fell by a factor of 10 to 27, from 0.272 to 0.010 at $\alpha = 0.2$ and from 0.178 to 0.033 at $\alpha = 0.05$. All numbers below use the pooled protocol. That almost all of the run-to-run scatter in the earlier stages was optimizer basin selection rather than anything physical is, we think, the single most useful practical lesson in this work.

*Direction.* The predicted displacement direction is essentially exact. The cosine between the measured offset change $\Delta\epsilon_0$ and $-d_\alpha$, taken in the $W$ metric, has median 0.997, 0.999, 0.980 and 0.992 at $\alpha = 0.2$, 0.1, 0.05 and 0.025, with Fisher-combined null probabilities between $10^{-13}$ and $10^{-20}$ and not a single significantly anti-aligned point. We restrict this statistic to points where a local displacement model can apply at all—the gain must be resolved above its own floor, the CVaR arm must not have found a better mean optimum (it jumped basins rather than trading), and $|R^{\mathrm{raw}}|$ must exceed 0.005 so that there is a displacement to have a direction. All exclusions and the ungated statistics are reported in the Supplemental Material [30]. We note that a minimum-cosine criterion, which we had pre-registered, is the wrong statistic: $\cos = 0.53$ in three dimensions has null probability 0.47 and is uninformative rather than contradictory.

*Level.* Here the pre-registration failed and something better replaced it. We had asked for a through-origin slope in $[0.7, 1.3]$. The measured slope is

$$s_\alpha \;=\; 0.588 \pm 0.025 \tag{18}$$

across the four $\alpha$ levels, and $0.589 \pm 0.033$ across all eight $(\alpha, \mathrm{seed})$ cells, with Spearman $\rho$ between 0.85 and 0.97 (Table IV, Fig. 16). Because scatter in the dependent variable caps the attainable correlation rather than attenuating the slope, the right comparison for $\rho$ is against that ceiling: the ratios are 0.98, 0.97, 0.88 and 0.98. The predictor therefore *orders* the available gain almost perfectly and predicts its magnitude up to one multiplicative constant, and that constant is flat in $\alpha$, in $\kappa$ and in seed. Flatness in $\alpha$ is the informative part: a finite-training-tail effect would scale with $\alpha$, since the CVaR objective sees 64 training draws at $\alpha = 0.2$ and 8 at $\alpha = 0.025$, and it does not.

The constant has a visible mechanism. The optimizer travels only part of the predicted step: the ratio $\|\Delta\epsilon_0\|/\|t\, d_\alpha\|$ has median 0.50 (Fig. 16c). For a gain quadratic in the offset and peaking at the predicted step, travelling a fraction $f$ returns $f(2-f)$ of it, which is 0.75 at $f = 0.5$; the remainder is the geometry co-moving, since the predictor holds $A$ and $Q$ fixed while the control moves. We do not claim 0.588 is universal. We claim it is a single constant on this device family, measurable once, and that what a hardware experiment needs from a predictor—which way to move and whether the move is worth making—is the part that comes for free.

Two pre-registered tests remain untested rather than failed, and we say so plainly. The resolution floor $R_{\min}$ sits at $\sim 10^{-4}$ while every point on our $\kappa$ grid predicts a gain above $3 \times 10^{-3}$, so the below-floor null test (H4) never engaged; and the predicted boundary $\kappa_c \approx 0.0047$ lies below the smallest $\kappa$ we ran, 0.0253, so the observed boundary is a censoring bound and not a measurement. The $\kappa^2$ law is at least self-consistent with it—at $\alpha = 0.05$, $C = 4.53$ gives $R = 1 \times 10^{-4}$ at $\kappa = 0.0047$, which is the floor—but that is arithmetic. Locating the boundary needs a predictor-only sweep down to $\kappa \approx 0.003$, which is cheap and which we did not run.

### M. What suppresses the gain: alignment, not rarity

Section IV K found that a rare catastrophic channel adds weight to the tail without adding leverage, and we generalized that to a claim about rarity: any rare subpopulation dominating the tail should suppress the gain, because compensating it costs the overwhelming majority of ordinary draws. That generalization is wrong, and the experiment built to confirm it refuted it instead.

Define the bulk share of the tail loss,

$$\Lambda_\alpha \;=\; \frac{\sum_{k \in \mathrm{tail} \cap \mathrm{bulk}} w_k L_k}{\sum_{k \in \mathrm{tail}} w_k L_k}, \tag{19}$$

which for the mixture of Proposition 4 is $\alpha\rho_\alpha/(\beta + \alpha\rho_\alpha)$ to first order in $p/\alpha$. For the relaxation channel this is exactly the right quantity. Splicing the recorded $(p, \beta)$ of each Stage 5 cell onto its measured dephasing ensemble—the faithful reconstruction, since Stage 5 used the exact no-jump/one-jump decomposition rather than sampling jumps—we find $\Lambda_\alpha$ agreeing with the analytic bulk share to a median 0.24%, and, more usefully,

$$\frac{R^{\mathrm{mix}}_{\mathrm{pred}}}{R^{\mathrm{bulk}}_{\mathrm{pred}}} \;=\; \Lambda_\alpha \tag{20}$$

to a median 0.65% and a worst case of 3.0% over 40 cell–$\alpha$ pairs spanning $\beta$ from 0.10 to 3.3 and $\Lambda$ from 0.70 to 0.075 (Fig. 17a). Leverage does not merely correlate with the gain; it multiplies it.

The counterexample is a two-state fluctuator: rare like a jump, but producing a coherent phase error that the pulse can in principle reshape. Sweeping its rarity $p$ at fixed excursion, with the fluctuator installed in the simulator so that the optimizer, the losses and the propagators all see it, gives Fig. 17b. Over a 25-fold range in $p$, $\Lambda_{0.05}$ falls from 0.836 to 0.018—a factor of 46—and $R^{\mathrm{mix}}_{\mathrm{pred}}$ rises by 2.7, while the *measured* gain does not move: 0.0167, 0.0163, 0.0154, 0.0157, flat to 4%. Eq. (20) fails here by up to two orders of magnitude, and fails in the direction of predicted enhancement.

The resolution is that $\Lambda_\alpha$ is a proxy, exact in one limit, for the quantity that actually governs: the fraction of the

tail loss the control can reduce. A relaxation jump contributes a fixed loss that no pulse can move, so its entire contribution is irreducible and the reducible fraction is $\Lambda_\alpha$ exactly. A fluctuator that loads the same error coordinate the bulk already loads is as reducible as the bulk is—the cosine between the bulk-only and mixture offsets stays between 0.999 and 0.944 across the sweep—so the reducible fraction stays near unity whatever $\Lambda$ does, and the gain is unchanged however rare the channel is. Consistently, the measured gain tracks the *bulk-only* predictor across the sweep, at a ratio of 0.40, 0.39, 0.37, 0.34, while its ratio to the mixture predictor falls from 0.37 to 0.13.

Both of our earlier readings named one axis of a two-axis quantity. "Shapeable versus inert" is the Stage 5 reading and is right about the mechanism but silent about rarity; "bulk versus rare" is right about the weights and silent about reducibility. What the two arms together establish is that rarity alone does not suppress a tail gain, alignment does, and Eq. (20) holds in the inert limit. An interpolation of the form $\Lambda + (1-\Lambda)\cos^2$ reproduces the observed ratio drop across the rarity sweep to about 6%; on four points we record that as a conjecture and not a law.

For a hardware campaign this is the more useful statement of the two. The question to ask of a noise channel is not how rare it is or how heavy its worst realization is, but whether the correction it wants points the same way as the correction the bulk wants—and that angle is estimable from tail-conditional error coordinates, the same measurement the predictor of Sec. IV L already needs.

Two limits bound these claims. The rarity sweep leaves the mixture regime at $p = 0.1$, where $p/\alpha = 2$ and the "rare" branch is twice the tail; and the largest fluctuator excursions drive the two-coordinate fit quality to $R^2 = 0.97$ and 0.68, below which nothing computed from the reduced model is interpretable. Separately, the fluctuator arm's measured gains were obtained with a single multistart rather than the pooled protocol of Sec. IV L, so the flatness across the sweep, rather than any individual value, is what carries the conclusion.

## V. DISCUSSION

*What breaks the equivalence.* The theorem predicts the region in which a tail objective is redundant, and its hypothesis (A2), measured as $\kappa$, bounds that region. Of the four physical routes by which the single-scalar structure can fail, leakage did not break it in our model; both objectives suppress leakage to the thermal floor and the residual distributions remain ordered. Two channels on different axes can break it (Proposition 2), but only weakly in practice. Every large divergence has one cause: second-order error that survives optimization, which CVaR pre-compensates for the tail draws. It arises from large accumulated detuning phase and, less obviously, from correlated channels, whose cancellation of the linear error exposes the second-order term. The fourth route, non-Gaussian randomness, we can now report on: trajectory-level $T_1$ relaxation reshapes the loss distribution exactly as Proposition 4 says and yet does not change which pulse is selected, for a reason that has no analogue in $\kappa$ (Sec. IV K). Three further experiments sharpen this picture: at fixed $\kappa$ the gate time still predicts the gain, so $\kappa$ orders the divergence without explaining it, while the geometry-aware reduced model does explain it (Sec. IV F); extending the error coordinate to the leakage amplitudes brings qubit and transmon onto one threshold (Sec. IV G); and the mechanism survives changes of rotation angle, anharmonicity and pulse basis (Sec. IV H).

TABLE IV. Calibration of the measurement-only predictor, by tail level. Slopes are through the origin on 16 points; the attenuation ceiling is the largest correlation attainable given the replicate scatter in $R$, so $\rho$/ceiling is the fraction of explainable variance the predictor captures. The last row is the fraction of the predicted step the optimizer actually travels.

| $\alpha$ | 0.20 | 0.10 | 0.05 | 0.025 |
|---|---|---|---|---|
| through-origin slope $s_\alpha$ | 0.558 | 0.617 | 0.594 | 0.582 |
| Spearman $\rho$ | 0.968 | 0.959 | 0.853 | 0.968 |
| attenuation ceiling | 0.990 | 0.989 | 0.968 | 0.991 |
| $\rho$ / ceiling | 0.978 | 0.970 | 0.881 | 0.977 |
| replicate s.d. of $R$ | 0.0101 | 0.0164 | 0.0328 | 0.0192 |
| median cos | 0.9970 | 0.9930 | 0.9760 | 0.9827 |
| median $\|\Delta\epsilon_0\|/\|\text{step}\|$ | 0.561 | 0.514 | 0.503 | 0.526 |

*Deciding in advance, and what actually suppresses the gain.* Two results change how the rest of this work would be used on a device. The first is that the gain is predictable from measurements on the mean-optimal pulse alone: the tail-conditional displacement gives the direction almost exactly—median cosine above 0.98 at every $\alpha$—and the magnitude up to a single multiplicative constant $0.588 \pm 0.025$ that is flat in $\alpha$, $\kappa$ and seed, with the predictor capturing 88–98% of the explainable variance in the measured gain (Sec. IV L). One decides whether to spend shots on a tail objective without first spending them. The second corrects the reading we gave of the relaxation result. It is not rarity that suppresses the gain but misalignment: the ratio of the mixture prediction to the bulk-only prediction equals the bulk share of the tail loss to 0.65% for the control-inert jump channel, and fails by two orders of magnitude for a fluctuator whose preferred correction points the same way as the bulk's, whose rarity can be swept 25-fold with no effect on the measured gain at all (Sec. IV M). The question to ask of a noise channel is whether the pulse can move it, not how rare it is. A third observation is methodological but was the largest single effect we measured: most of the run-to-run scatter in these experiments was optimizer basin selection, and equalizing the restart budget between the two objectives cut the replicate scatter in

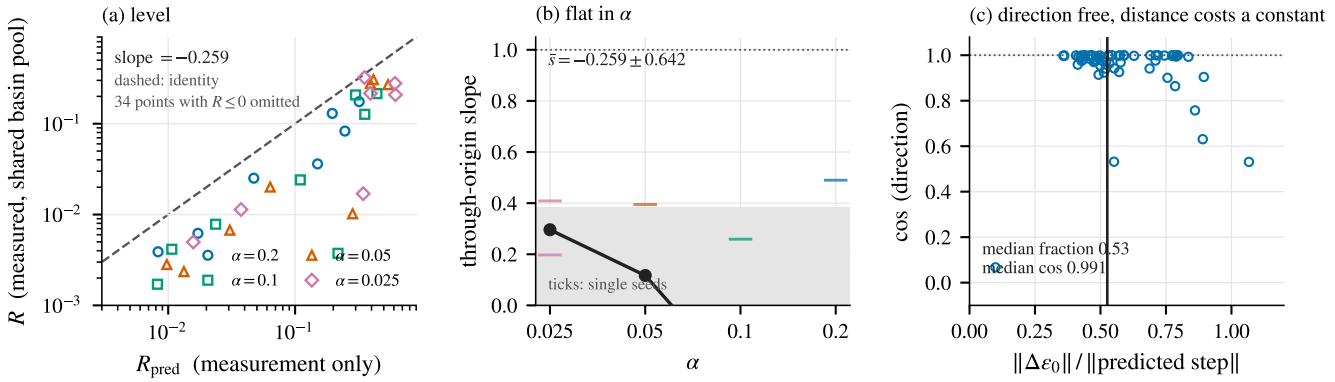



FIG. 16. A measurement-only predictor of the tail gain. (a) Measured gain against $R_{\text{pred}}$ for the 16 points of the $\kappa$ sweep at four $\alpha$ levels; the solid line is the fitted through-origin slope 0.588 and the dashed line is the identity. (b) That slope against $\alpha$, with single-seed values as ticks and the band showing $\pm 1$ standard deviation across $\alpha$; it is flat, which rules out a finite-training-tail explanation. (c) The mechanism behind the constant: the direction is predicted almost exactly (median cos $> 0.98$) while the optimizer travels a median 0.50 of the predicted step.

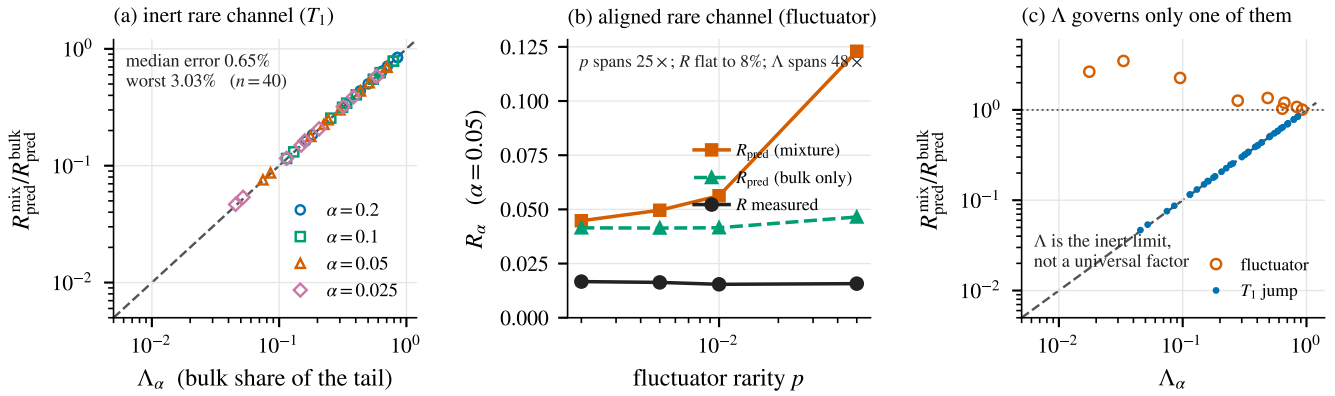



FIG. 17. Alignment, not rarity, governs the suppression. (a) For the control-inert relaxation channel, the ratio of the mixture prediction to the bulk-only prediction equals the bulk share $\Lambda_\alpha$ of the tail loss over two decades, with median error 0.65% across 40 cell–$\alpha$ pairs. (b) For a two-state fluctuator that loads the same coordinate as the bulk, sweeping its rarity over 25× moves $\Lambda$ by 46× and the mixture prediction by 2.7× while the measured gain stays flat to 4% and tracks the bulk-only prediction instead. (c) Both arms on one axis: $\Lambda$ is the inert limit of the reducible fraction, not a universal suppression factor.

the gain by a factor of 10 to 27.

*A test for practitioners.* The results reduce the decision whether to use a tail objective to a measurement on the mean-optimal control. (i) Optimize the mean as usual. (ii) Simulate $u_{\text{mean}}$ on a few hundred noise draws and extract the error rotation vectors. (iii) Fit Eq. (10) and compute $\kappa$ and, preferably, $R_\alpha^{\text{model}}$. (iv) Compute the coefficient $C$ of Eq. (13) from the same fit and compare $C\kappa^2$ with the resolution $R_{\min}$ you can actually measure. If $C\kappa^2 \lesssim R_{\min}$ — for one dominant noise direction and $\alpha = 0.05$, if $\kappa \lesssim 0.32\sqrt{R_{\min}}$ — stop: a tail objective cannot improve the tail beyond estimation noise and can make it slightly worse out of sample. Use $\kappa$ alone only as a screen: it is a safe one, since no control with small $\kappa$ diverged anywhere in this study, but it is not sufficient, because at fixed $\kappa$ the attainable gain still varies by an order of magnitude with the error geometry that $C$ and $R^{\text{model}}$ keep and $\kappa$ discards (Sec. IV F). (v) Otherwise optimize CVaR warm-started from $u_{\text{mean}}$, with $\alpha K \gtrsim 5$ training draws in the tail, and expect a gain of at most $R_\alpha^{\text{model}}$. For a single frequency-noise channel, step (iii) can be replaced by the accumulated phase variance, known before any optimization: below $\sigma_\phi^2 \approx 0.05$ CVaR is not worth running. With several correlated channels $\kappa$ must be measured, because the optimizer itself decides how much linear error remains. In our data the threshold $R_{0.05}^{\text{model}} > 0.08$ classifies significant divergence correctly at 89% of points, and $\kappa > 0.23$ at 82%.

One diagnostic in this list costs nothing and needs no fit. The tail-to-mean ratio of the mean-optimal control, already available from step (ii), gives $m_{\text{eff}}$ by inverting Eq. (6). A ratio near $r_\alpha$ means a single noise direction

reaches the gate, which is the regime the closed-form coefficient of Table I describes; a ratio well below it means several do, the tail is already lighter than the worst case, and both the amplification a tail objective exploits and the fitted $C$ are smaller (Sec. IV I). Dephasing that is fast on the gate scale is the clearest instance. The same inversion works whatever raises $m_{\mathrm{eff}}$: memory and channel geometry give residuals from Eq. (6) that no test distinguishes (Sec. IV J).

*One caveat the screen does not cover.* Steps (iii)–(iv) measure how nonlinear the residual error is. They do not measure how much of a channel the control can actually move, and Sec. IV K shows the two can come apart completely: a relaxation channel carrying 98% of the mean loss, sitting above every other draw in the tail and weighted a further 3.58 times by the objective, produced no change in the selected pulse, because the gate requirement pins its probability to within 1.4% and its cost is fixed by the target unitary. Where a heavy channel is suspected, the cheap follow-up is to compare two controls that differ in mean loss and ask how much that channel's contribution differs between them. If the answer is a percent, no reweighting of it will change the answer, however far into the tail it sits.

*Relation to previous work.* Most robust-control methods target the mean or a low-order expansion of the error: ensemble GRAPE and sampling-based learning control average the fidelity over noise [4–6], sequential convex programming bounds it [7], and composite pulses, filter functions and universally robust control cancel its leading-order sensitivities [9–11]. Our results state when these approaches are already tail-optimal: whenever the residual error is linear in Gaussian noise, a mean-optimal control is optimal for every monotone risk measure. They also explain why cancelling the linear error, the aim of all these methods, is what makes a tail objective matter, since it raises $\kappa$. The exponential-utility controls of Ref. [19], which differed from average-fidelity controls, are covered by Corollary 1, which gives a sufficient condition under which no such difference is possible. On the risk side, the equivalence is an instance of a general property of stochastic orders [22, 23] and parallels the coincidence of mean–variance and mean–CVaR portfolios for elliptical returns [14, 20]. What is specific to quantum gates is the periodic loss, handled here with an explicit residual; the rotation geometry, which breaks the ordering through majorization; and the second-order structure of the error on a driven qubit, which breaks it through pre-compensation.

*Physical reading.* A CVaR-optimized gate beyond the boundary is deliberately miscalibrated. On a perfectly calibrated day it performs worse than the mean-optimized gate, and it performs better on days with large drift. Whether this is desirable depends on the error-correction layer: if logical failure is dominated by the worst rounds, the tail is the relevant quantity; if drifts are tracked and recalibrated frequently, the mean may be. The present results show that the trade exists only at large $\kappa$ and can be sized in advance from the mean control alone.

*Methodological note.* The comparison of two optimization objectives is fragile. In the course of this work, seven distinct pipeline errors produced publishable-looking but false conclusions, among them a regularizer that manufactured $R_{\mathrm{tail}} \approx 0.3$ at the leakage onset, and a comparator asymmetry that made CVaR look arbitrarily bad ($R_{\mathrm{tail}} = -3.6$ in one cell against $+0.11$ at equal search effort) (Sec. S3.D). Search effort is the central issue: a mean optimizer stuck in a poor basin makes any alternative look good, and extra search on one side only makes the other look bad. Neither the equal-budget nor the best-found comparison is safe alone. We recommend the warm-start audit, the reverse warm start and the reporting of both comparisons as standard practice for such studies.

*Limitations and outlook.* The noise is Gaussian, and $T_1$ enters only as a Lindblad channel: the one failure route of Sec. II C that the theorem cannot address is untested. The colored study covers one gate time and the two-level model, and at $\tau_c \leq 0.3\,T_g$ its two-coordinate reduction of the noise breaks down ($R^2$ as low as 0.010), so $C$ and $R^{\mathrm{model}}$ are unreliable exactly where $R_{\mathrm{tail}} \approx 0$; with two seeds per cell the reproducible spread of $R_{\mathrm{tail}}$ there is only 0.015, which caps any rank correlation at 0.60 and makes that stage a test of consistency rather than of discrimination. Part (iv) of Proposition 3 is numerical, not proved, and degrades as $\alpha \to 0$. The reduced model optimizes only the offset, and its coefficient $C$ is read from a least-squares fit of the error cloud, so it degrades where that fit does. The controlled design fixes $\kappa$ of the control the calibration optimizes, not of the one the run finally selects, which is larger by a median factor 1.08 and up to 1.74, so its rows overlap; with two seeds per cell, a knob effect below roughly a factor of two per cell would not be resolved. The generality test covers the sine basis well and the piecewise-constant basis poorly, because the latter optimizes unreliably at matched parameter count. Natural next steps are trajectory-level $T_1$ statistics; non-Gaussian noise, where assumption (A1) rather than (A2) is at stake; two-qubit gates, where the error geometry has more directions than the offset can cancel; and a hardware test of the predicted crossing of the loss distributions on a pulse-level superconducting processor.

## VI. CONCLUSION

When the gate error is linear in Gaussian noise, the error distributions of all controls are rescaled copies of one shape, and the mean-optimal control is optimal for CVaR and for every other law-invariant monotone risk measure. For single-qubit gates under quasi-static detuning this holds while the accumulated phase variance stays below about 0.05. There, over hundreds of candidate controls per setting, the two objectives never disagree on a well-

separated pair, and every control shows the predicted tail-to-mean ratio. Leakage does not change this. Beyond the boundary, the second-order error of the mean-optimal control, measured by $\kappa$, decides whether and by how much a tail objective helps: the attainable gain is quadratic in $\kappa$ with a coefficient we give in closed form, so the threshold below which a tail objective is redundant follows from the theory rather than from the data, and a parameter-free reduced model predicts the size of the gain above it. What the tail objective then finds is a deliberately miscalibrated gate that pre-compensates the second-order error of the worst noise draws. A controlled experiment at fixed $\kappa$ shows that this scalar orders the divergence but does not by itself explain it: the error geometry that the reduced model keeps is what makes a gate at one duration different from a gate at another with the same $\kappa$. Giving the noise memory does not change any of this — across two decades of correlation time the comparison depends on the correlation only through that same coefficient — but it does change the shape of the tail, by spreading the error over more noise directions. The tail-to-mean ratio measures how many, along a curve that contains the single-channel and two-channel ratios as its first two points — and rotating the noise between two channels, a mechanism with no time structure at all, puts the ratio on that same curve within the scatter of the first route. A rare catastrophic channel is the one case where the shape of the tail and the choice of pulse come apart. Adding $T_1$ relaxation at the trajectory level moves the tail-to-mean ratio from 5.6 to almost the ceiling $1/\alpha$, in quantitative agreement with a closed form, while leaving the selected pulse untouched: the gate requirement fixes both the probability of relaxation and its cost to within about a percent, so an objective that weights that channel more heavily has nothing to act on. Selection responds to how much of a channel the control can move, not to how much of the loss it carries. A fluctuator sharpens the point: it is rare like a jump but loads the same error coordinate the bulk does, and sweeping its rarity over a factor of twenty-five changes the measured gain by four percent while the bulk share of the tail moves by a factor of forty-six. What suppresses a tail gain is misalignment between the correction the rare channel wants and the one the bulk wants, and the bulk share is an exact proxy for it only in the inert limit that relaxation happens to occupy.

Finally, none of this has to be discovered by running both optimizations. The tail-conditional displacement of the error coordinates under the mean-optimal control predicts which way a tail objective would move the gate—median cosine above 0.98—and how much it would gain, up to one multiplicative constant near 0.59 that is flat in the tail level, in $\kappa$ and in seed. Since the displacement, the metric and the reachable subspace are all measurable on hardware, the decision of whether a tail objective is worth its shots can be made before any of them are spent. Robust gates can therefore be designed for the mean, and tested for $\kappa$, for that alignment, and for the predicted gain, before a tail objective is worth its cost.

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*Data availability.* The code and the data underlying the figures are available from the author upon reasonable request.

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